Dissertação - João Paixão, Iago Leal

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1 Introduction

Category theory has often been regarded as the mathematics of mathematics [Lan98]. The capacity to lift particular concepts into a more general and abstract space has demonstrated, time and again, that not just the natural world, but all of mathematics seems to be governed by universal constructions. The investigation of such universal concepts allows us to see what a particular view could not, and to realize the intrinsic connections and shared nature of what was once thought to be separate.

One such field, that appears throughout mathematics yet stands apart, is optimization. Optimal thinking is present in every mathematical construct in which one seeks the "ideal" object in some criterion. This notion of "ideal" often refers to the simplest object that meets certain criteria. Prior work has already characterised the internal optimization structure within categorical constructs [Kad26], but such results are still isolated; a unified treatment is still to be formalized.

This thesis is motivated by the observation that many important mathematical structures used in optimization can be understood within a single compositional framework [SS24; Han+24]. Instead of treating relations, convex functions, quadratic forms, polyhedra, and linear programs as isolated objects, this work shows that they can all be interpreted as morphisms in categories whose composition is given by elimination or minimization. In this way, optimization itself becomes an algebraic process and an universal setting across mathematics.

The development follows a natural progression. Classical binary relations appear as Boolean-valued morphisms, weighted relations arise through Lawvere quantales, convex bifunctions introduce infimal composition, quadratic relations capture least-squares and energy minimization, polyhedral relations describe systems of inequalities, and finally linear programming extends the framework to optimization problems with decision variables and constraints. Each level enlarges the expressive power of the previous one while preserving compositional structure.

Figure 1 illustrates the path from the theory of interacting hopf algebras [BSZ17] to more expressive languages presented in this thesis, one generator at a time. It also illustrates how adding a specific class of "measurement" generators lifts the category from the binary relations category (indicator functions) into the one of weighted relations (bifunctions/optimization problems).

This expressivity becomes especially apparent when analyzed through the language of string diagrams [JS91; Sel10; PZ23]. Originating as a rigorous graphical language for monoidal categories, string diagrams provide an intuitive yet mathematically precise syntax in which the sequential and parallel composition of morphisms correspond directly to vertical and horizontal juxtaposition of diagrams [JS91; Lan98]. Building on this foundation, a rich family of *diagrammatic* or *graphical calculi* has emerged, offering complete axiomatisations of various algebraic structures purely through equational diagram rewriting [Sel10; BSZ17]. These calculi have proved remarkably versatile, finding applications across a wide range of fields: in *quantum computing* [CD11]; *electronics* [BF18]; *linear algebra* [BSZ17; Bon+19; Fre+25]; and in *optimisation and probability* [Ste+24; SS21; BDS21].Throughout this work we explore how, step by step, one can enrich the already known theories for linear [BSZ17] algebra by adding one new generator at a time, along with a particular set of axioms, and derive even richer structures for convex problems [SS24].

We start by demonstrating how the addition of the new generator of Graphical Quadratic Algebra (**GQA**) [Ste+24] expands the expressivity of previous theories by acting as a sort of measurement device along the diagram. This new mechanism enables diagrams to represent the broader class of weighted relations, rather than the more restricted class of binary relations. From the optimization perspective, this is what plays the role of the objective function

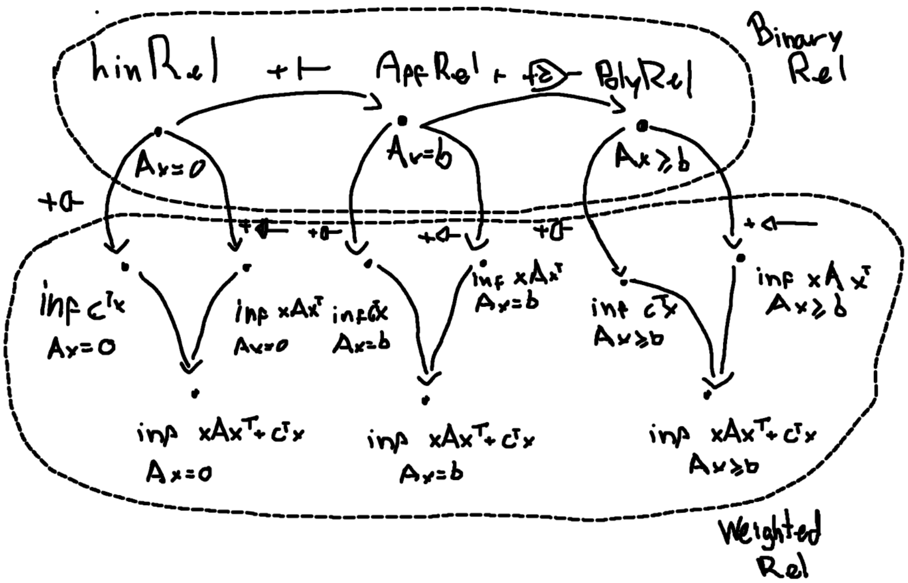


Figure 1: This image represent the category of subcategories of Rel; We circle two class of subcategories, namely those that are contained in the Binary Relations categories and those embed in Weighted Relations category. Observe how morphism between objects (class of relations) are the addition of a single generator (along with some axioms), and that there are particular generators which are essential binary relations (those that are horizontal) that keeps objects in the subclass of binary relations, but there are others(the vertical morphisms), that resembles an output or measurement, that lift the relations to those on the Weighted Relations class

in an optimization problem. More precisely, the language of **GQA** expresses minimization problems of the ℓ_2 norm of vectors over some affine subspace, i.e. ,

$$\inf_x \{\|Ax + b\|^2 \,:\, x \in Cx = d\}$$

Next, we extend the Symmetric Monoidal Theory of Graphical Affine Algebra (**GAA**) [Bon+19] into the even more general theory of Symmetric Inequality Monoidal Theory. This is achieved by adding the inequality generator $- \geq -$. Geometrically, this allows our algebra to go beyond affine spaces and start representing polyhedral figures as linear inequality constraints in optimization.

$$\{|x \in P\} = \begin{cases} 0 & if\ Ax + b \geq 0 \\ +\infty & \text{otherwise} \end{cases}$$

Inspired by these observations, we make our central contribution in the final section. The introduction of graphical algebra for parametric linear programming. In this setting, a linear program is no longer viewed only as a numerical problem to be solved, but as a morphism that can be composed with other optimization blocks. This gives a symbolic language in which the elimination of internal variables corresponds directly to categorical composition, allowing optimization problems to be manipulated diagrammatically before computation.

In particular, every piecewise affine convex function can be represented as the value function of a linear program [Roc70], which means the Graphical Linear Programming ((GLP)) theory provides a syntax for a broad class of nonsmooth convex models that appear in optimization, control, economics, and machine learning. In Section 6 we show the diagrammatical equivalence of this two classes of objects and how diagrammatically they are not only denotationally equivalent but in fact instances of the same normal form. Thus, our diagrammatical syntax blurs the distinction between expressing LP programs as optimization problems and piecewise convex functions deriving a canonical normal form.

More broadly, the thesis suggests that graphical methods can play for optimization the same role that graphical linear algebra plays for matrices: providing normal forms, symbolic rewriting rules, and completeness results within a rigorous categorical setting. This opens the possibility of using diagrammatic reasoning not only for proofs, but also for designing compositional optimization systems and formal symbolic calculi for convex programming.

Thesis overview In Section 2 we introduce the algebraic and categorical playground on which the rest of the work is built, present the initial notions of quantales and relations, and how they play an interchangeable role in a categorical view. Next, in Section 3, is introduced basic application of this theory to the vast field of convex optimization [SS24; Kad26], showing how many of its important constructs fall within such a categorical framework.

In Sections 4 and 5 is presented two already-known graphical theories, namely Graphical Quadratic Algebra [Ste+24] and Polyhedral Algebra [BDS21], where some of the known proofs are recalled and refined.

Finally, we define and prove the completeness of an entirely new Graphical Algebra dedicated to expressing parametric Linear Programs. This work represents a first step toward a complete diagrammatic calculus for convex optimization.

2 Ground definitions for relations

In this section we explore the theory of relations. While much of mathematical theory is built on functional thinking, we argue that most of mathematics actually has a relational nature. Category theory is the natural language for this purpose. It is precisely built on top of this notion, in which everything one needs to know about an object is determined by its morphisms.

To begin we define the most basic kind of relation:

Definition 1 (Binary Relations)**.** *Let X and Y be sets. A* binary relation *R from X to Y is a subset*

$$R \subseteq X \times Y.$$

Equivalently, it can be identified with its characteristic function

$$\chi_R : X \times Y \to \{0, +\infty\},$$

defined by

$$\chi_R(x, y) = \begin{cases} 0 & \text{if } (x, y) \in R, \\ +\infty & \text{otherwise.} \end{cases}$$

A binary relation specifies wheather x is related to y. Observe that one can think of a relation in two different ways that will prove useful later. First, as a set of pairs, where every relation is just a pair $(x, y) \in R$, or as a map that has x, y as inputs and outputs 0 if they are related, and $+\infty$ if they are not the choice of ∞ is arbitrary and will be justified later. Normal choices here are actually $\{true, false\}$ or $\{0, 1\}$)

This is a general construct as the sets X and Y can be anything. We can, for instance, ask for further structures on such sets and obtain differents kinds of relations.

Definition 2 (Linear/Affine Relations)**.** *Let X and Y be vector spaces. A* linear relation *from X to Y is a linear subspace*

$$R \subseteq X \times Y.$$

An affine relation *is an affine subspace of $X \times Y$.*

Equivalently, a linear relation can be described as the graph of a possibly multivalued linear operator, i.e.,

$$R(x) := \{y \in Y \mid (x, y) \in R\}.$$

or, its characteristic function defined by

$$\chi_R : X \times Y \to \{0, +\infty\},$$

$$\chi_R(x,y) = \begin{cases} 0 & \text{if } (x,y) \in R, \\ +\infty & \text{otherwise.} \end{cases}$$

Thus, a linear/affine relation is a binary relation where the set R is a subspace

Notice, however, that we have been restricting the relations between x and y to be a case of "they are, or they are not related". In fact, we might be interested in adding degrees to such a relation, so that we could say that x and y are "more" or "less" related. This leads us to the well-known notion of weighted relations [Law73].

Definition 3 (Weighted Relations)**.** *Let X and Y be any two sets and let $\overline{\mathbb{R}}^+ := [0, +\infty]$. A* weighted relation *from X to Y is a set*

$$R \subseteq X \times Y \times \overline{\mathbb{R}}^+$$

such that for every $(x,y) \in X \times Y$ there exists exactly one $r \in \overline{\mathbb{R}}^+$ *satisfying*

$$(x,y,r) \in R.$$

Equivalently, we may define the characteristic function

$$\chi_R : X \times Y \to \overline{\mathbb{R}}^+$$

by

$$\chi_R(x,y) = r \quad \text{if and only if } (x,y,r) \in R.$$

and the codomain is not restricted to a binary set anymore. In fact, one could choose any other set instead of the $\overline{\mathbb{R}}^+$ to be the codomain of the characteristic function as long as it satisfy some properties. In fact, the whole structure of a relation is further generalized by the notion of a quantale.

Definition 4 (Quantale)**.** *A* quantale *is a structure $(Q, \leq, \otimes, e)$ such that:*

- *$(Q, \leq)$ is a complete lattice, i.e., every subset $A \subseteq Q$ admits a join $\bigvee A$ and a meet $\bigwedge A$;*
- *$(Q, \otimes, e)$ is a monoid;*
- *Multiplication distributes over arbitrary join in each variable:*

$$a \otimes \Big(\bigvee_{i\in I} b_i\Big) = \bigvee_{i\in I}(a \otimes b_i), \qquad \Big(\bigvee_{i\in I} a_i\Big) \otimes b = \bigvee_{i\in I}(a_i \otimes b)$$

  for all families $\{a_i\}_{i\in I}, \{b_i\}_{i\in I} \subseteq Q$.

If $\otimes$ is commutative, the quantale is called commutative*. If $e = \top$ (the top element of the lattice), it is called* integral*.*

Given a quantale Q, we can define the notion of Q-valued relation [Coe+18].

Definition 5 (Generalized Relations)**.** *A* generalized relation *(or Q-valued relation) from X to Y, where Q is a quantale, is a set*

$$R \subseteq X \times Y \times Q$$

such that for every $(x, y) \in X \times Y$ there exists exactly one *$q \in Q$ satisfying*

$$(x, y, q) \in R.$$

Its characteristic function is

$$\chi_R : X \times Y \to Q$$

defined by

$$\chi_R(x, y) = q \quad \text{if and only if } (x, y, q) \in R.$$

Now we are ready to build our previous relations as particular cases of this more general framework.

Example 1. *The notion of a Q-valued relation recovers familiar concepts for particular choices of quantale.*

(1) Truth quantale. *Let $Q = \{0, 1\}$ with the usual order $0 \leq 1$, join given by $\vee$, and monoidal product $\otimes = \wedge$ with unit $e = 1$. This is the (Boolean) truth quantale.*

A Q-valued relation is then a function

$$R : X \times Y \to \{0, 1\}.$$

Such a function is precisely the characteristic function of an ordinary binary relation

$$\widetilde{R} \subseteq X \times Y, \qquad (x, y) \in \widetilde{R} \iff R(x, y) = 1.$$

Thus Q-valued relations coincide with classical binary relations.

(2) Lawvere quantale. *Let $Q = \overline{\mathbb{R}}^+ = [0, +\infty]$ ordered by the* reverse *of the usual order (so that $a \leq b$ iff $a \geq b$ in the usual order), with monoidal product given by addition*

$$a \otimes b := a + b,$$

and unit $e = 0$. Joins are then given by infima in the usual order.

This structure is the Lawvere quantale. A Q-valued relation

$$R : X \times Y \to \overline{\mathbb{R}}^+$$

is precisely a weighted relation assigning a (possibly infinite) cost to each pair (x, y).

Hence generalized relations unify ordinary binary relations and weighted relations as special cases corresponding to particular quantales.

Note that linear and affine relations are not recovered by this construction, since they demand extra structure on the sets X and Y, speciffically that R be a linear or affine subspace of $X \times Y$. They are, nonetheless, fully embedded in the framework of binary relations.

Within this framework, we propose an extension of the Lawvere Quantale and weighted relations called the Extended-Lawvere Quantale and the Cost/Reward Relations, respectively. To our knowledge, this definition does not appear elsewhere in the literature.

Definition 6 (Optimization/Extended Lawvere Quantale)**.** *Let*

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

The optimization quantale *is the structure*

$$(\overline{\mathbb{R}}, \geq, \otimes, e)$$

defined as follows:

- *The order $\geq$ is the usual order on the extended real line. With this order, arbitrary joins are given by the usual infima:*

$$\bigvee A = \inf A.$$

- *The monoidal product is addition:*

$$a \otimes b := a + b,$$

  where addition is extended in the usual way (with $a + (+\infty) = +\infty$ and $a + (-\infty) = -\infty$ whenever defined).

- *The unit element is*

$$e := 0.$$

With these operations, $(\overline{\mathbb{R}}, \geq, +, 0)$ is a commutative quantale. Joins are ordinary infima, and addition distributes over arbitrary joins.

Definition 7 (Optimization (Cost/Reward) Relations)**.** *Let X and Y be sets. An* optimization relation *from X to Y is an $\overline{\mathbb{R}}$-valued relation, i.e., a function*

$$R : X \times Y \to \overline{\mathbb{R}}.$$

Equivalently, it is a subset

$$R \subseteq X \times Y \times \overline{\mathbb{R}}$$

such that for every $(x, y) \in X \times Y$ there exists exactly one $r \in \overline{\mathbb{R}}$ satisfying

$$(x, y, r) \in R.$$

The value $R(x, y)$ is interpreted as the objective value (cost or reward) associated to the pair (x, y), with $+\infty$ and $-\infty$ representing extremal infeasibility or unboundedness.

Observe how this is essentially the weighted relations extended along the negative values. The importance of it is that we want to be able to express optimization problems that might be unfeasible ($+\infty$) or unbounded ($-\infty$), so we need to add both the top and bottom element.

All of this admits a suitable and natural categorical construction. Specifically, from any quantale Q one can not only define Q-relations, but also construct an entire category of them. This category is the natural setting for discussing the central concept of this framework: the composition of relations.

The following lemma shows how to build such a category and establishes that this category also has a very important monoidal structure behind it. Particularly, the category is a compact closed symmetric monoidal category [Lan98]:

Proposition 1 (Category $\mathbf{Rel}(Q)$)**.** *For a quantale Q, the Q-relations form a category $\boldsymbol{Rel}(Q)$ with composition and identities*

$$(S \circ R)(a, c) = \bigvee_b R(a, b) \otimes S(b, c) \quad 1_A(a, b) = \bigvee \{k | a = b\}$$

If Q is a commutative quantale, then $\boldsymbol{Rel}(Q)$ carries a symmetric monoidal structure, with the categorical monoidal product $\bigotimes_{\mathcal{C}}$ on objects given by the Cartesian product of sets and the action on relations given for $R : A \to C$ and $S : B \to D$ by

$$(R \otimes_{\mathcal{C}} S)(a, b, c, d) = R(a, c) \otimes S(b, d)$$

The singleton set is the monoidal unit, and the category is compact closed w.r.t the monoidal structure

Proof. We verify the categorical structure.

(1) Well-defined composition. Let $R : A \to B$ and $S : B \to C$ be Q-relations. For each $(a, c) \in A \times C$, the set

$$\{R(a, b) \otimes S(b, c) \mid b \in B\} \subseteq Q$$

admits a supremum since Q is a complete lattice. Hence $(S \circ R)(a, c)$ is well-defined as an element of Q.

(2) Associativity. Let $R : A \to B$, $S : B \to C$, and $T : C \to D$. For $a \in A$, $d \in D$,

$$((T \circ S) \circ R)(a, d) = \bigvee_c \Big(\bigvee_b R(a, b) \otimes S(b, c) \Big) \otimes T(c, d).$$

Since $\otimes$ distributes over arbitrary suprema,

$$= \bigvee_{b,c} R(a, b) \otimes S(b, c) \otimes T(c, d).$$

Similarly,

$$(T \circ (S \circ R))(a, d) = \bigvee_b R(a, b) \otimes \Big(\bigvee_c S(b, c) \otimes T(c, d) \Big) = \bigvee_{b,c} R(a, b) \otimes S(b, c) \otimes T(c, d).$$

Thus composition is associative.

(3) Identities. For each set A, define

$$1_A(a,b) = \begin{cases} e & \text{if } a = b, \\ \perp & \text{otherwise,} \end{cases}$$

where $\perp = \bigvee \varnothing$ is the bottom element of Q.

Then for $R : A \to B$,

$$(R \circ 1_A)(a,b) = \bigvee_{a'} 1_A(a,a') \otimes R(a',b).$$

All terms vanish except when $a' = a$, so

$$= e \otimes R(a,b) = R(a,b).$$

Similarly, $1_B \circ R = R$. Hence identities exist.

(4) Monoidal structure (commutative case). Assume Q is commutative. Define the tensor on objects by Cartesian product:

$$A \otimes_{\mathcal{C}} B := A \times B.$$

For relations $R : A \to C$ and $S : B \to D$, define

$$(R \otimes_{\mathcal{C}} S)((a,b),(c,d)) = R(a,c) \otimes S(b,d).$$

Functoriality follows from associativity and commutativity of $\otimes$:

$$(S_1 \otimes S_2) \circ (R_1 \otimes R_2) = (S_1 \circ R_1) \otimes (S_2 \circ R_2).$$

The singleton set 1 is a unit since $A \times 1 \cong A$.

Symmetry is induced by the swap map

$$\sigma_{A,B}((a,b),(b',a')) = \begin{cases} e & \text{if } a = a',\ b = b', \\ \perp & \text{otherwise.} \end{cases}$$

(5) Compact closure. For each set A, define unit and counit relations

$$\eta : 1 \to A \times A, \qquad \epsilon : A \times A \to 1$$

by

$$\eta(*,(a,a')) = \begin{cases} e & a = a', \\ \perp & \text{otherwise,} \end{cases} \qquad \epsilon((a,a'),*) = \begin{cases} e & a = a', \\ \perp & \text{otherwise.} \end{cases}$$

A direct computation using the definition of composition shows that the triangle identities hold:

$$(1_A \otimes \epsilon) \circ (\eta \otimes 1_A) = 1_A, \qquad (\epsilon \otimes 1_A) \circ (1_A \otimes \eta) = 1_A.$$

Thus every object is self-dual, and $\mathbf{Rel}(Q)$ is compact closed. □

A curious fact about relations is that they actually encode the notion of matrices and matrix multiplication. This serves as motivation for what will later become the diagrammatical theory of linear algebra. The following remark expands on it showing that every morphism in $\mathbf{Rel}(Q)$ corresponds to a matrix with entries in Q, and that composition is exactly matrix multiplication.

Remark 1. *A Q-valued relation $R : X \times Y \to Q$ is nothing but a Q-*valued matrix *indexed by $X \times Y$: one simply reads off the entry at position (x, y) as $\chi_R(x, y) \in Q$. That is, a generalized relation is precisely an element of $Q^{X\times Y}$.*

When X and Y are finite sets, say $|X| = m$ and $|Y| = n$, this becomes completely explicit: a Q-valued relation is exactly an $m \times n$ matrix

$$M_R \;=\; \big[\chi_R(x_i,\, y_j)\big]_{\substack{1\le i\le m\\ 1\le j\le n}} \;\in\; Q^{m\times n},$$

whose (i, j)-entry records the Q-value assigned to the pair (x_i, y_j).

Moreover, composition of Q-valued relations corresponds precisely to matrix multiplication in Q*. Given a second Q-valued relation $S : Y \times Z \to Q$, their composite $S \circ R : X \times Z \to Q$ is defined by*

$$\chi_{S\circ R}(x, z) \;=\; \bigvee_{y\in Y} \chi_R(x, y) \otimes \chi_S(y, z),$$

which, in matrix notation, reads

$$M_{S\circ R} \;=\; M_R \otimes M_S, \qquad (M_R \otimes M_S)_{ij} \;=\; \bigvee_{k=1}^{n} (M_R)_{ik} \otimes (M_S)_{kj}.$$

This is exactly the usual row-times-column matrix product, but with ordinary addition replaced by the join $\bigvee$ and ordinary multiplication replaced by the monoidal product $\otimes$ of Q. The two axioms of a quantale — that $\otimes$ distributes over arbitrary joins — are precisely what guarantees that this multiplication is associative and well-defined, so that Q-valued matrices form a category under this composition.

To illustrate these new concepts, we look at some of our previous relations now built as categories.

Example 2 (Category of Boolean relations)**.** *Let*

$$Q = (\{0, 1\}, \le, \otimes, e)$$

be the Boolean quantale where:

- $0 \le 1$*;*
- $\otimes = \wedge$ *(logical conjunction);*
- $e = 1$*;*

- *joins are given by* $\vee$.

A Q*-valued relation* $R : A \to B$ *is a function*

$$R : A \times B \to \{0, 1\},$$

which is precisely the characteristic function of a subset

$$\widetilde{R} \subseteq A \times B.$$

The composition formula in $\mathbf{Rel}(Q)$ *becomes*

$$(S \circ R)(a, c) = \bigvee_b R(a, b) \wedge S(b, c).$$

Since joins are logical disjunction, this reduces to

$$(S \circ R)(a, c) = 1 \quad \iff \quad \exists b \in B \text{ such that } R(a, b) = 1 \text{ and } S(b, c) = 1.$$

Thus composition coincides exactly with the usual composition of binary relations.

Since the Boolean quantale is commutative, $\mathbf{Rel}(Q)$ *carries a symmetric monoidal structure:*

- *On objects:*
$$A \otimes B := A \times B.$$

- *On morphisms:*

$$(R \otimes S)((a, b), (c, d)) = R(a, c) \wedge S(b, d).$$

The singleton set is the monoidal unit. With this structure, $\mathbf{Rel}(Q)$ *is precisely the classical symmetric monoidal compact closed category* $\mathbf{Rel}$.

Example 3 (Category of weighted relations)**.** *Let*

$$Q = (\mathbb{R} \cup \{+\infty\}, \leq_Q, \otimes, e)$$

be the Lawvere quantale where:

- $a \leq_Q b \iff a \geq b$ *in the usual order;*

- $\otimes = +$*;*

- $e = 0$*;*

- *joins are given by infima in the usual order.*

A Q-valued relation $R : A \to B$ is a function

$$R : A \times B \to \mathbb{R} \cup \{+\infty\},$$

assigning a cost (or weight) to each pair (a, b).

The composition formula becomes

$$(S \circ R)(a, c) = \inf_{b \in B} \big(R(a, b) + S(b, c) \big).$$

Thus composition is given by infimal convolution *(or shortest-path composition).*

Since addition is commutative, the Lawvere quantale is commutative, and $\mathbf{Rel}(Q)$ *carries a symmetric monoidal structure:*

- *On objects:*
$$A \otimes B := A \times B.$$

- *On morphisms:*

$$(R \otimes S)((a, b), (c, d)) = R(a, c) + S(b, d).$$

The singleton set is the monoidal unit. With this tensor product, $\mathbf{Rel}(Q)$ *is a symmetric monoidal compact closed category of weighted relations.*

In particular:

- *If A, B, C are finite sets, relations may be viewed as matrices with entries in* $\mathbb{R} \cup \{+\infty\}$*.*

- *Composition is matrix multiplication in the* $(\min, +)$*-semiring.*

Hence $\mathbf{Rel}(Q)$ *is the category of weighted relations, which underlies dynamic programming, tropical algebra, and shortest-path problems.*

Those examples illustrate how naturally the quantale structure supports categorical constructions: composition of Q-valued relations corresponds to matrix multiplication driven by $\otimes$, and the tensor product of relations is governed by the join $\bigvee$, both operations coming directly from the quantale axioms.

The resulting categories are symmetric monoidal, and it is precisely this monoidal backbone that will support the algebraic structure we aim to build. A key piece of that structure is a *Frobenius monoid* on each object: a monoid (μ, η) and a comonoid (δ, ε) on the same object, satisfying the Frobenius law

$$(\mathrm{id} \otimes \mu) \circ (\delta \otimes \mathrm{id}) \;=\; \delta \circ \mu \;=\; (\mu \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \delta).$$

A symmetric monoidal category in which every object carries such a structure, compatibly with the tensor product, is called a *hypergraph category* [FS19].

Hypergraph categories are related to, but strictly more general than, compact closed categories [Sel10]. In a compact closed category every object A has a dual A^* with unit $\eta_A : I \to A^* \otimes A$ and counit $\varepsilon_A : A \otimes A^* \to I$ satisfying the snake equations; a Frobenius structure does not require the existence of such a dual. Conversely, if a compact closed category is such that every object is self-dual ($A \cong A^*$) and the induced cups and caps satisfy the Frobenius law, then it is a hypergraph category. Thus compact closed categories with self-dual objects form a special class of hypergraph categories, and it is the later, more general, notion that we shall adopt as our ambient framework.

Remark 2. *If Q is commutative, then* $\mathbf{Rel}(Q)$ *is a hypergraph category and admits a commutative Frobenius structure on every object. Diagrammatically, we denote the additive and co-additive structure as*

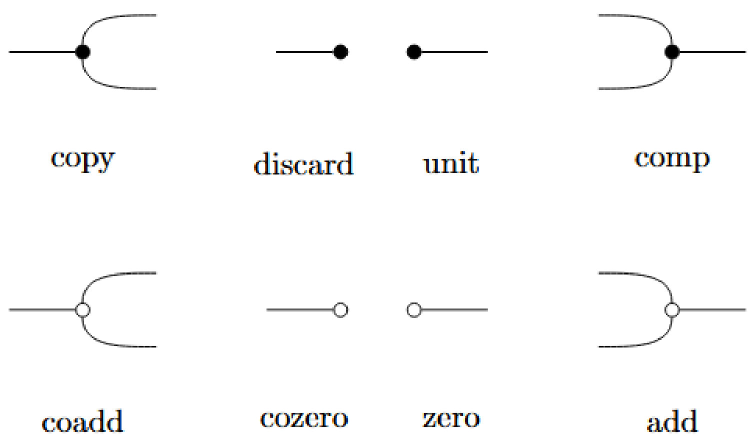
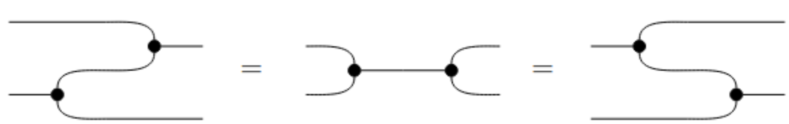


Figure 2: Frobernius additive(black) and co-additive(white) structure

Subject to the Frobernius Law (both colors)



Proof. Just a check that all laws do hold for any $\mathbf{Rel}(Q)$ category ☐

These constructions provide exactly the semantic foundation for the algebra we aim to build. In the same way that semantics is understood in linguistics and programming language theory [Win93], the semantics of a formal system assigns meaning to its expressions. More precisely, while syntax consists of the tokens and rules for forming and combining expressions, semantics is the interpretation assigned to each such expression — in our case, an optimization problem living in $\mathbf{Rel}(Q)$.

As outlined in the introduction, the central goal of this work is to show that optimization emerges naturally from algebraic and categorical structure. The key mechanism enabling this is the infimal composition present in the definition of the Costs/Rewards category: composing two relations automatically performs variable elimination, which is precisely the fundamental operation underlying optimization.

To make this precise, we need both a semantics — provided by the categorical framework developed above — and a syntax: a rigorous, compositional language for expressing optimization problems symbolically. The map between those is structure-preserving map, a functor, from the syntactic layer into the semantic one.

The syntactic layer will be formalized as a Symmetric Monoidal Theory (SMT) [BSZ17], which is a presentation of a category by generators and equations, whose terms respect the compositional structure of a symmetric monoidal category. This is the standard tool for building graphical languages in the style of Graphical Linear Algebra, and it is the framework within which our completeness results will be stated and proved.

Definition 8 (Symmetric Monoidal Theory)**.** *A SMT is a pair* (Σ, E)*, such that,* Σ *is a signature of operators* o *that has arity* m *and coarity* n

Σ*-terms are defined inductively as the smallest collection of typed expressions* $t : m \to n$ *such that:*

1. *(Generators) If* $o \in \Sigma$ *with* $o : m \to n$*, then* o *is a* Σ*-term of type* $m \to n$*.*
2. *(Identity) For every* $n \in \mathbb{N}$*, there is a term*

   $$\mathrm{id}_n : n \to n.$$

3. *(Composition) If*

   $$a : m \to n \quad \text{and} \quad b : n \to k$$

   are Σ*-terms, then*

   $$a ; b : m \to k$$

   is a Σ*-term.*
4. *(Tensor product) If*

   $$a : m_1 \to n_1 \quad \text{and} \quad b : m_2 \to n_2$$

   are Σ*-terms, then*

   $$a \otimes b : m_1 + m_2 \to n_1 + n_2$$

   is a Σ*-term.*
5. *(Symmetry) For all* $m, n \in \mathbb{N}$*, there is a term*

   $$\sigma_{m,n} : m + n \to n + m.$$

and E be a set of equations over the $\Sigma - terms$.

Often, we will think of elements $o : m \to n \in \Sigma$ as generators and represent them as string diagrams. Being the $\Sigma - terms$ the results of composing those generators horizontally and vertically inductively. The set E as a collection of axiomatic equations over these string diagrams.

The categorical counterpart of these terms is the free category generated by the SMT.

Definition 9 (The Free category of SMT). *The $FreeP_{(\Sigma,E)}$ is the category freely generated by (Σ, E) where the objects are natural numbers, morphisms are $\Sigma - terms$ quotiented by E, with composition and tensor product inherited from the symmetric monoidal structure. $FreeP_{(\Sigma,E)}$ is in fact a PROP*[1].

The $\text{FreeP}_{(\Sigma,E)}$, also denoted as $P\Sigma_{/E}$, [2] will be our **syntax** responsible for expressing our underlying semantics. For it to be considered a "useful" syntax we expect that it satisfies some properties, particularly we want it to be complete, sound and expressive.

Without those, our algebra would be empty since we would have no assurances that it actually says anything meaningful about we desire to say.

Definition 10. *We will say that a SMT axiomatizes or represents a specific category, usually called the semantic category, when the following diagram commutes:*

$$\begin{tikzcd}
\mathsf{P}\Sigma_{/E} \arrow[r, "\cong"] \arrow[dr, "s"] & \mathsf{Im}([\![\cdot]\!]) \arrow[d, hook, "i"] \\
\mathbf{Syn} = \mathsf{P}\Sigma \arrow[u, "q"] \arrow[r, "{[\![\cdot]\!]}"'] & \mathbf{Sem}
\end{tikzcd}$$

where: $P\Sigma$ is the category where objects are natural numbers and morphisms are $\Sigma - terms$ from $n \to m$; $P\Sigma_{/E} = FreeP_{(\Sigma,E)}$ is the $P\Sigma$ category with the morphisms quotiented by E and there is then a prop morphism $q : P\Sigma \to P\Sigma_{/E}$ witnessing this quotient; $\boldsymbol{Sem}$ is our semantic category (the Cost/Reward relations for example); and s is a full and faithful prop morphism. Another way to say the same is to demand two things out of the SMT, namely:

- *Soundness: We say that $P\Sigma_{/E}$ is sound if $c \overset{E}{=} d$*[3] *implies $[c] = [d]$.*
- *Completeness: We say that $P\Sigma_{/E}$ is complete if $[c] = [d]$ implies $c \overset{E}{=} d$*
- *Expressivity: We say that $P\Sigma_{/E}$ is expressive if for every morphism c in $\boldsymbol{Sem}$ there exists $d \in P\Sigma_{/E}$ such that $[d] = c$*

[1] For the definition of a prop see [Lan98]

[2] During the work we will use FreeP_S and S, for whatever syntax S may be, interchangeably.

[3] $\overset{E}{=}$ means that this equality is provable by the equations in E

Those three properties capture the idea that everything we can prove in the syntax, is a holds in the semantics (soundness), everything that holds in the semantics is provable in the syntax (completeness) and that every morphism in the semantics has a syntactic representative (expressivity). Moreover, these properties denote equivalent characteristics about the interpration map, both categorically and functionally. Below we depict a small table defining the equivalence between the desired properties and their equivalent statement:

Table 1: Equivalence of properties for the interpretation map $[\![-]\!] : P\Sigma_{/E} \to \mathbf{Sem}$

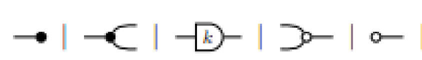
Set theory	**Category theory**	**Logic (SMT)**
Total	Functor	Soundness: $c \overset{E}{=} d \;\Rightarrow\; [\![c]\!] = [\![d]\!]$
Deterministic	Well-defined map	(Functionality of the interpretation)
Injective	Faithful	Completeness: $[\![c]\!] = [\![d]\!] \;\Rightarrow\; c \overset{E}{=} d$
Surjective	Full	Expressivity: $\forall f \in \mathbf{Sem},\ \exists d \in P\Sigma_{/E}$ s.t. $[\![d]\!] = f$

Now we introduce two of the foundational theories for Graphical Algebra, that will serve as a foundation for the following sections, namely, Graphical Linear Algebra [PRS22; BSZ17] and Graphical Affine Algebra [Bon+19], we omit their definitions and proofs and refer to the original sources.

However, we do provide some notation on how to talk about those during the course of the work. Notice that, their semantics was already introduced when we talked about Linear/Affine relations.

Definition 11. *We will call* $\boldsymbol{GLA}$ *the presentation of the Graphical Linear Algebra SMT, and, in the same manner,* $\boldsymbol{GAA}$ *the SMT of Graphical Affine Algebra.*

Definition 12. *We also denote with a* $\overrightarrow{./.}$ *when we want to talk about the algebra generated without the frobernius structure. For example* $\overrightarrow{\boldsymbol{GLA}}$ *is the algebra generated by*



without their symmetric

And the exactly opposite we denote as $\overleftarrow{\boldsymbol{GLA}}$.

3 Composability of Convex Problems

Having established the necessary categorical and algebraic tools, we proceed to their application in convex analysis. In this section we introduce a initial categorical framework for convex optimization [SS24], showing how the fundamental operations of convex analysis — infimal convolution, conjugation, and variable elimination — arise naturally as categorical constructions. This gives the first concrete evidence that optimization is not merely a collection of isolated techniques, but a coherent algebraic theory with deep compositional structure.

The progression of the section is as follows. We begin with the general category of bifunctions, which serves as the broadest setting, and progressively restrict to convex bifunctions, where the analytical structure is richest. We then introduce the Legendre–Fenchel transform and show that bifunction adjunction defines a contravariant functor between the convex and concave bifunction categories. These results lay the groundwork for the graphical theories developed in the following sections: Graphical Quadratic Algebra and Polyhedral Algebra will both arise as full subcategories of **CxBiFn**, each corresponding to a class of bifunctions with additional algebraic structure.

Later, we will see the particular applications of such definitions and results. For now, however, we begin by defining the broadest category in the hierarchy, which will generalize most of the later work: the category of bifunctions.

Definition 13 (Bifunction Category)**.** *The category* **BiFunc** *is defined as follows:*

- *Objects are sets.*
- *A morphism* $F : A \to B$ *is a function*

$$F : A \times B \to \overline{\mathbb{R}}.$$

- *Given* $F : A \to B$ *and* $G : B \to C$*, their composition* $G \circ F : A \to C$ *is defined pointwise by*

$$(G \circ F)(a, c) := \inf_{b \in B} \big(F(a, b) + G(b, c)\big).$$

- *The identity morphism on an object* A *is the function*

$$1_A : A \times A \to \overline{\mathbb{R}}$$

given by

$$1_A(a, a') = \begin{cases} 0 & \textit{if } a = a', \\ +\infty & \textit{otherwise.} \end{cases}$$

If we equip **BiFunc** *with a monoidal structure:*

- *The tensor product on objects is the Cartesian product*

$$A \otimes B := A \times B.$$

- *For morphisms* $F : A \to C$ *and* $G : B \to D$*, their tensor product*

$$F \otimes G : A \times B \to C \times D$$

  is defined by

$$(F \otimes G)((a,b),(c,d)) := F(a,c) + G(b,d).$$

The singleton set serves as the monoidal unit.

Remark 3. *The category* **BiFunc** *is isomorphic to the category* ***RelQ*** *for a suitable choice of* Q*.*

Consider the quantale

$$Q = (\overline{\mathbb{R}}, \leq_Q, \otimes, e)$$

where:

- $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$*;*
- *the order is reversed with respect to the usual order,*

$$a \leq_Q b \quad \Longleftrightarrow \quad a \geq b \quad \textit{(usual order)};$$

- *the monoidal product is addition,*

$$a \otimes b := a + b;$$

- *the unit is* $e = 0$*.*

In this quantale, arbitrary joins are given by infima in the usual order:

$$\bigvee\nolimits_Q A = \inf A.$$

Now a Q*-valued relation* $R : A \to B$ *is precisely a function*

$$R : A \times B \to \overline{\mathbb{R}},$$

i.e. a bifunction. Moreover, the composition formula in $\mathbf{Rel}(Q)$ *becomes*

$$(S \circ R)(a,c) = \bigvee_b R(a,b) \otimes S(b,c) = \inf_b \big(R(a,b) + S(b,c)\big),$$

which is exactly the composition law in **BiFunc***.*

Therefore, **BiFunc** *is (strictly) the same category as* $\mathbf{Rel}(Q)$ *for the Lawvere-type quantale* $(\overline{\mathbb{R}}, \geq, +, 0)$*.*

Bifunctions are fundamental objects in convex analysis [Roc70]. They provide a unified language for expressing objective functions, feasibility constraints, and duality simultaneously. The following proposition confirms that the framework is conservative: classical set-theoretic relations embed faithfully into this richer setting.

Proposition 2. *The category of Binary Relations, i.e. the Boolean-relation is embed as a full subcategory of BiFunc.*

Proof. Just consider the functor $F : \mathbf{Rel}(B) \to \mathbf{BiFunc}$ that takes every morphism $R : X \to Y$, which is a subset of $X \times Y$, and maps it into the characteristic function $\chi_R : X \times Y \to \{0, +\infty\}$, also known as the indicator function $1_R(x, y)$, or in bracket notation $\{|(x, y) \in R|\}$ $\square$

Having established the general framework, we now restrict our attention to the convex setting, which is the natural home for optimization. The key insight is that convexity is preserved under infimal composition, so the convex bifunctions form a subcategory of **BiFunc** that is closed under the categorical operations.

Definition 14 (Convex Bifunction Category **CxBiFn**). *Let*

$$Q := (\overline{\mathbb{R}}, \leq_Q, \otimes, e)$$

be the extended-Lawvere quantale where:

- $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$*;*
- $a \leq_Q b \iff a \geq b$ *in the usual order;*
- $a \otimes b := a + b$*;*
- $e := 0$.

The category **CxBiFn** *is defined as follows:*

- ***Objects:*** *finite-dimensional real vector spaces.*
- ***Morphisms:*** *A morphism* $F : A \to B$ *is a function*

$$F : A \times B \to \overline{\mathbb{R}}$$

  satisfying:

  - F *is proper (not identically* $+\infty$*);*
  - F *is convex as a function on the product space* $A \times B$*;*
  - F *is lower semicontinuous.*

- ***Composition:*** *Given* $F : A \to B$ *and* $G : B \to C$*, their composite*

$$G \circ F : A \to C$$

  is defined by infimal convolution:

$$(G \circ F)(a, c) := \inf_{b \in B} \big(F(a, b) + G(b, c)\big).$$

- ***Identity:*** *For each object A, the identity morphism*

$$1_A : A \to A$$

is given by

$$1_A(a,a') = \begin{cases} 0 & \textit{if } a = a', \\ +\infty & \textit{otherwise.} \end{cases}$$

- *Monoidal product*
  - *On objects:*

$$A \otimes B := A \times B.$$

  - *On morphisms: For $F : A \to C$ and $G : B \to D$,*

$$(F \otimes G)((a,b),(c,d)) := F(a,c) + G(b,d).$$

  - *The monoidal unit is the zero-dimensional vector space $\{0\}$.*

Remark 4. *One may define the analogous category* **CvBiFn** *which is the category of concave bi functions. For doing so, one should only change the composition from inf to sup, this is:*

$$(G \circ F)(a,c) := \sup_{b \in B} \big(F(a,b) + G(b,c)\big).$$

Every result for **CxBiFn** *will hold, in a equivalent way, to the category* **CvBiFn**

Remark 5. *Every finite-dimensional real vector space V is linearly isomorphic to $\mathbb{R}^n$, where $n = \dim V$.*

Hence, one may restrict the objects of **CxBiFn** *to the family*

$$\mathrm{Ob}(\mathbf{CxBiFn}) = \{\mathbb{R}^n \mid n \in \mathbb{N}\}.$$

With this choice, **CxBiFn** *becomes a skeletal category, since $\mathbb{R}^n \cong \mathbb{R}^m$ implies $n = m$.*

Also, if one considers the isomorphism $n \leftrightarrow \mathbb{R}^n$, **CxBiFn** *becomes a prop, this can be made strict by defining the monoidal product on objects as sum.*

$$m \otimes n = m + n$$

which is isomorphic to

$$\mathbb{R}^m \otimes \mathbb{R}^m = \mathbb{R}^m \times \mathbb{R}^m = \mathbb{R}^{m+n}$$

Observe that we have constructed a category whose morphisms are interpreted as convex optimization problems. Since every bifunction $F : \mathbb{R}^m \times \mathbb{R}^n \to \overline{\mathbb{R}}$ can be written as

$$F(x, y) = \inf_z z \quad s.t\ z = F(x, y)$$

This observation is central to the thesis. It shows that there's some sort of duality between relations and optimization. For a single morphism, however, this remains imprecise. The compositional structure yields more substantive insights.

$$(S \circ R)(x, z) = \inf_y \{R(x, y) + S(y, z)\}$$

Which shows that relational composition is precisely variable elimination: composing two relations automatically marginalises over the intermediate variable y by minimising the combined cost. In the binary case this reduces to the existential question of whether a mediating element exists; in the weighted case it asks for the *optimal* such element. Optimization is therefore not imposed on the categorical structure from outside — it emerges from composition itself.

Convex theory also has a very nice duality theory which is of particular interest in algebraic constructs such as this one. We now introduce the central duality operation of convex analysis The Legendre–Fenchel transform.

Definition 15 (Legendre-Fenchel Transform)**.** *For a convex function $f : \mathbb{R}^n \to \overline{\mathbb{R}}$, its convex conjugate $f^* : \mathbb{R}^n \to \overline{\mathbb{R}}$ is defined as:*

$$f^*(x^*) = \sup_x \{\langle x^*, x \rangle - f(x)\}.$$

Geometrically, f^* encodes information about all affine functions majorised by f: each supporting hyperplane of the epigraph of f corresponds to a point where f^* is finite [Roc70]. This duality plays a central role across many fields in mathematics and physics such as convex optimization and duality [BV04], probability theory and large deviations [DZ10], optimal transport [Vil09] and Hamiltonian mechanics.

The Legendre–Fenchel transform plays a role in convex optimization analogous to that of transposition in linear algebra: it exchanges the roles of primal and dual variables, and, as we shall see, it interacts functorially with categorical composition [Wil15].

Such transformation can be extended to bifunctions providing the equivalent of the Legendre–Fenchel transform for morphisms in **CvBiFn**, providing a notion of duality that is compatible with categorical composition.

Definition 16 (Adjoint of convex bifunctions)**.** *The adjoint of a convex bifunction F is the concave bifunction F^* defined by*

$$F^*(y^*, x^*) = \inf_{x,y} \{F(x, y) + \langle x^*, x \rangle - \langle y^*, y \rangle\}$$

Note that F^* is a closed concave bifunction and $F^{**} = cl(F)$, also, if F is closed, then:

$$F^{**} = F$$

The following theorem is the central result of this section. It shows that bifunction adjunction is not merely an analytic operation, but a categorical one: it defines a contravariant functor between **CxBiFn** and **CvBiFn**. This is the bifunction analogue of the fact that matrix transposition reverses composition, and it will be the algebraic basis for the duality diagrams appearing in the graphical theories of the next sections.

Theorem 1 (Functoriality of the bifunction adjoint)**.** *Let $F : \mathbb{R}^m \times \mathbb{R}^n \to \overline{\mathbb{R}}$ and $G : \mathbb{R}^n \times \mathbb{R}^p \to \overline{\mathbb{R}}$ be proper, convex, and lower semicontinuous functions.*

Define their infimal composition

$$(G \overset{\rightharpoonup}{\circ} F)(x,z) := \inf_{y \in \mathbb{R}^n} \big\{ F(x,y) + G(y,z) \big\}.$$

Assume that a standard qualification condition holds, e.g.

$$\operatorname{ri}(\operatorname{dom} F) \cap \operatorname{ri}(\operatorname{dom} G) \neq \varnothing$$

in the y-variable.

Then

$$(G \overset{\rightharpoonup}{\circ} F)^* = F^* \overset{\rightharpoondown}{\circ} G^*,$$

where the supremal composition is defined by

$$(F^* \overset{\rightharpoondown}{\circ} G^*)(z^*, x^*) := \sup_{y^* \in \mathbb{R}^n} \big\{ F^*(y^*, x^*) + G^*(z^*, y^*) \big\}.$$

Hence it defines a contravariant functor between the category **CxBiFn** *and the category* **CvBiFn***.*

Proof. Let $F : \mathbb{R}^m \times \mathbb{R}^n \to \overline{\mathbb{R}}$ and $G : \mathbb{R}^n \times \mathbb{R}^p \to \overline{\mathbb{R}}$ be proper, convex, and lower semicontinuous. Define

$$H(x,z) := (G \overset{\rightharpoonup}{\circ} F)(x,z) = \inf_{y \in \mathbb{R}^n} \{ F(x,y) + G(y,z) \}.$$

By definition,

$$H^*(z^*, x^*) = \inf_{x,z} \{ H(x,z) + \langle x^*, x \rangle - \langle z^*, z \rangle \}.$$

Substituting the definition of H,

$$H^*(z^*, x^*) = \inf_{x,z} \inf_{y} \Big\{ F(x,y) + G(y,z) + \langle x^*, x \rangle - \langle z^*, z \rangle \Big\}.$$

Since the infimum is taken over independent variables,

$$H^*(z^*, x^*) = \inf_{x,y,z} \Big\{ F(x,y) + G(y,z) + \langle x^*, x \rangle - \langle z^*, z \rangle \Big\}.$$

Insert the identity

$$0 = \sup_{y^*} \{ \langle y^*, y \rangle - \langle y^*, y \rangle \}$$

to prepare for dualization:

$$H^*(z^*, x^*) = \inf_{x,y,z} \sup_{y^*} \Big\{ F(x, y) + \langle x^*, x\rangle - \langle y^*, y\rangle + G(y, z) + \langle y^*, y\rangle - \langle z^*, z\rangle \Big\}.$$

Under the qualification condition

$$\mathrm{ri}(\mathrm{dom}\, F) \cap \mathrm{ri}(\mathrm{dom}\, G) \neq \varnothing \quad \text{in the } y\text{-variable},$$

Fenchel–Rockafellar duality (or a standard minimax theorem) allows the interchange of inf and sup:

$$H^*(z^*, x^*) = \sup_{y^*} \inf_{x,y,z} \Big\{ F(x, y) + \langle x^*, x\rangle - \langle y^*, y\rangle + G(y, z) + \langle y^*, y\rangle - \langle z^*, z\rangle \Big\}.$$

The expression separates into two independent infima:

$$= \sup_{y^*} \Big\{ \inf_{x,y} \big(F(x, y) + \langle x^*, x\rangle - \langle y^*, y\rangle \big) + \inf_{y,z} \big(G(y, z) + \langle y^*, y\rangle - \langle z^*, z\rangle \big) \Big\}.$$

By definition of the bifunction adjoint,

$$F^*(y^*, x^*) = \inf_{x,y} \big\{ F(x, y) + \langle x^*, x\rangle - \langle y^*, y\rangle \big\},$$

and

$$G^*(z^*, y^*) = \inf_{y,z} \big\{ G(y, z) + \langle y^*, y\rangle - \langle z^*, z\rangle \big\}.$$

Hence

$$H^*(z^*, x^*) = \sup_{y^*} \big\{ F^*(y^*, x^*) + G^*(z^*, y^*) \big\}.$$

Therefore,

$$(G \overset{\rightharpoonup}{\circ} F)^* = F^* \overset{\rightharpoondown}{\circ} G^*,$$

where

$$(F^* \overset{\rightharpoondown}{\circ} G^*)(z^*, x^*) = \sup_{y^*} \{ F^*(y^*, x^*) + G^*(z^*, y^*) \}.$$

Finally, since the Fenchel conjugation is involutive on proper, convex, lower semicontinuous functions, we also have $H^{**} = H$, and the adjoint reverses composition. This completes the proof. □

The results of this section establish **CxBiFn** as a well-behaved categorical setting for convex optimization. The two special classes of morphisms — quadratic bifunctions and polyhedral bifunctions — each inherit this structure while admitting a finite, explicit representation. It is precisely this finiteness that makes a graphical algebra possible: in both cases, morphisms can be encoded as diagrams composed from a finite set of generators, and the equational theory of the diagrams corresponds exactly to the equivalences between optimization problems. The following two sections make this precise.

4 Quadratic Relations

Among all convex bifunctions, quadratic ones are particularly well-structured: they are the simplest class rich enough to express least-squares problems, energy minimisation, and Gaussian inference, yet structured enough to admit a finite diagrammatic presentation. This section develops the Graphical Quadratic Algebra (GQA) [Ste+24], the first of the two graphical theories on which our linear programming algebra will be built. We begin with the necessary preliminaries on quadratic functions, then define the symmetric monoidal theory whose morphisms are non-negative quadratic optimization problems — the non-negativity condition ensuring that infimal composition is always well-posed. We then present the soundness and completeness results for this theory. The section closes by deriving several known results about quadratic problems purely through string diagram manipulation, demonstrating that the graphical syntax is not merely a notational convenience but a genuine calculus in which optimization arguments can be carried out.

4.1 Preliminary results about quadratic functions

Our main interest are non-negative quadratic functions. The motivation for this unusual choice is that we want to avoid unbounded problems remain on convex settings, since negative quadratic functions may be concave. Formally, this means that the category of partial non-negative quadratic functions embeds into the positive Lawvere-quantale.

Next, we rigorously define the notion of a non-negative quadratic function along with other relevant equivalence results.

Definition 17. *A partial quadratic function is a function*

$$f : \mathbb{R}^{n+m} \to [0, +\infty]$$

of the form

$$f(x, y) = \langle x, Ax \rangle + \langle b, x \rangle + c + \{|x \in M|\}$$

where A is a symmetric matrix, M is an affine subspace, $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$ and $\langle -, - \rangle$ denotes the usual inner product on the reals.

Definition 18. *An elementary convex partial quadratic function h is a function that can be written in the diagonal form $h(x) = \sum_i \lambda_i x_i^2$ for $\lambda_i \geq 0$. It is partial because $h(x) = \infty$ whener $\lambda_i = +\infty$ $x_i \neq 0$ for some i.*

Proposition 3. *The following statements are equivalent*

- *f is a non-negative partial quadratic function*
- *f can be written in the form $f(x) = h(A(x - a)) + c$ where A is an invertible matrix, $c \geq 0$ and h an elementary convex partial quadratic function*
- *f can be written as $f(x) = \inf\{||y||^2 : (x, y) \in M\}$ where M is an affine subspace*

Proof of Proposition 2. **(1) ⇒ (2).** Using an affine change of coordinates, we may assume that the domain of f is the subspace

$$M = \mathbb{R}^m \times \{0\}.$$

Define the restricted function

$$\tilde{f}(x_1, \dots, x_m) = f(x_1, \dots, x_m, 0, \dots, 0).$$

Then $\tilde{f}$ is a total nonnegative quadratic function. Hence, by the characterisation of total nonnegative quadratic functions, there exist an invertible matrix A, a vector a, a constant $c \geq 0$, and an elementary convex quadratic function h such that

$$\tilde{f}(x_1, \dots, x_m) = h\big(A((x_1, \dots, x_m) - a)\big) + c.$$

Returning to the original coordinates, we obtain

$$f(x) = h\big(A((x_1, \dots, x_m) - a)\big) + \infty \cdot x_{m+1} + \cdots + \infty \cdot x_n + c,$$

which is of the required form.

(2) ⇒ (3). Under the affine coordinate change

$$y = A(x - a),$$

it suffices to consider functions of the form

$$f(y) = h(y) + c,$$

where h is an elementary quadratic function. Up to a permutation of coordinates, we may write

$$h(y_1, y_2, y_3) = \|y_1\|_2^2 + \{\, y_3 = 0 \,\},$$

where $\{\, y_3 = 0 \,\}$ denotes the indicator function of the subspace $\{y_3 = 0\}$ (equal to 0 if $y_3 = 0$ and $+\infty$ otherwise).

Define the affine relation

$$M = \Big\{ \big((y_1, y_2, y_3), (y_1, \sqrt{c})\big) \;\Big|\; y_3 = 0 \Big\}.$$

Then, for every x,

$$f(x) = \inf\{\|z\|_2 : (x, z) \in M\}.$$

(3) ⇒ (1). Let $M \subset \mathbb{R}^n \times \mathbb{R}^m$ be an affine relation. Then there exists a vector subspace $D \subset \mathbb{R}^m$ and an affine map $g : \mathbb{R}^n \to D^\perp$ such that, for each x,

$$M_x = \{y : (x, y) \in M\} = \begin{cases} g(x) + D, & x \in \pi_X M, \\ \varnothing, & \text{otherwise}, \end{cases}$$

where $\pi_X M$ denotes the projection of M onto $\mathbb{R}^n$.

Hence,

$$\inf\{\|y\|_2 : y \in M_x\} = \|g(x)\|_2 + \{\, x \in \pi_X M \,\}.$$

This is clearly a nonnegative partial quadratic function. □

As it is stated above, a non-negative partial quadratic function has a very convenient algebraic structure that can be expressed both with and without explicit constrains.

This formulation reveals one of the deepest conceptual payoffs of the relational framework. In the standard convex analysis view [BV04], incorporating a constraint $y \in M_x$ into an objective function via its indicator

$$\inf\{\|y\|_2 \,:\, y \in M_x\} = \inf_y \left\{\|y\|_2 + \mathbf{1}_M(y)\right\}$$

is a familiar technique, closely related to the method of Lagrangian multipliers. In the relational setting, however, this operation acquires a precise categorical meaning: binary relations, identified with their $\{0, +\infty\}$-valued characteristic functions, represent the *feasibility constraints* of an optimization problem, while weighted relations represent its *objective function.* Composing the two — adding the indicator to the cost — is precisely categorical composition in **BiFunc**. Constraints and objectives are therefore not fundamentally different kinds of objects: they are relations valued in different quantales, unified by the same compositional structure.

This correspondence is made precise by the following proposition, which establishes that non-negative partial quadratic functions are closed under the categorical composition of **CxBiFn**.

Proposition 4. *If $M \subseteq \mathbb{R}^n \times \mathbb{R}^m$ is an affine subspace, and g is a non-negative partial quadratic function, then so is $f(x) = \inf\{g(y) \mid (x, y) \in M\}$*

Proof. Let $g : \mathbb{R}^m \to [-\infty, \infty]$ be a nonnegative partial quadratic function and let $M \subseteq \mathbb{R}^m \times \mathbb{R}^n$ be an affine relation. Using the representation of Proposition 16 and an affine change of coordinates, it suffices to consider the case where g is an elementary function. Hence we may assume that g is given by a constrained minimization problem of the form

$$f(x) = \inf\left\{\|y_1\|_2^2 + \{\, y_3 = 0 \,\} \mid (x, (y_1, y_2, y_3)) \in M\right\}.$$

Define the affine relation

$$M' = \{(x, y_1) \mid (x, (y_1, y_2, 0)) \in M\}.$$

Since M is affine, it follows directly that M' is also an affine relation. Moreover, the constraint $\{\, y_3 = 0 \,\}$ has been incorporated into the definition of M'.

Therefore,

$$f(x) = \inf\left\{\|y\|_2^2 \mid (x, y) \in M'\right\},$$

which is of the form characterizing partial quadratic functions. Hence f is a partial quadratic function.

□

The above result is the closure property that makes graphical theory well-founded: the composition of a quadratic objective with any number of affine constraints remains a quadratic problem. In other words, we can compose a weighted relation with as many affine relations as we want. This is equivalent of adding affine constrains to the problem.

Categorically, it states that non-negative partial quadratic functions form a subcategory of **CxBiFn** that is closed under infimal composition, and it is precisely this subcategory that the Graphical Quadratic Algebra axiomatises.

4.2 Graphical Algebra for quadratic problems

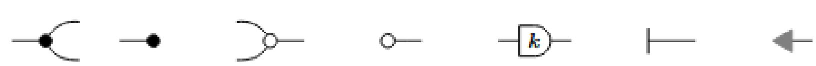
We now introduce the first axiomatic algebra of the thesis: the Graphical Quadratic Algebra (GQA) [Ste+24]. The goal of this diagrammatic calculus is to express and reason about optimization problems with non-negative quadratic objective functions and equality constraints. The canonical example of this class is the ordinary least-squares (OLS) estimator

$$\min_{x} \|Ax - b\|^2,$$

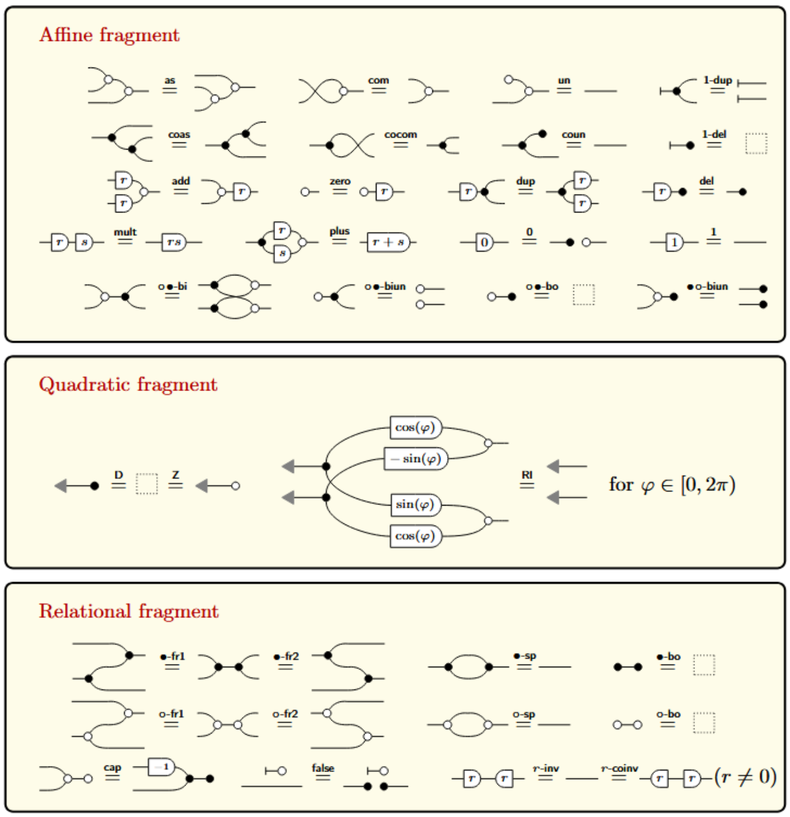
whose solution is given by the Moore–Penrose pseudoinverse $A^\dagger$. This is a problem of central importance in linear algebra, statistics, and control, and it serves as the principal worked example for GQA: later in this section we provide a purely diagrammatic argument showing that $A^\dagger$ solves the least-squares problem.

We proceed as follows. We first define the symmetric monoidal theory $(\Sigma_{\mathrm{GQA}}, E_{\mathrm{GQA}})$ by specifying its generators and equations. We then define the semantic category **QuadRel** of non-negative quadratic relations, and prove that **GQA** axiomatises it — that is, that the induced PROP morphism is full, faithful, and essentially surjective. The syntax is built on the following generators and equations E.

Definition 19 (Graphical Quadratic Theory)**.** *We call **GQA** the symmetric monoidal theory* (Σ, E)*, such that,* Σ *is depicted graphically as:*



Σ*-terms are defined by vertical and horizontal composition of terms with matching arity and coarity, and are modularized by* E*:*



The pair (Σ, E) *defines the symmetric monoidal theory generated by the operators* Σ *subject to the equations* E.

We will define the interpretation functor $[\![-]\!]$, *i.e., the semantics of each generator, as follows:*

$$[\![\;]\!]\left(x, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \{\!| x = x_1 = x_2 |\!\} \quad [\![\;]\!](x, ()) = 0 \quad [\![\;]\!]((), x) = 0$$

$$[\![\;]\!]\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y\right) = \{\!| y = x_1 + x_2 |\!\} \quad [\![\;]\!]((), x) = \{\!| x = 0 |\!\} \quad [\![\;]\!](x, ()) = \{\!| x = 0 |\!\}$$

$$[\![\;]\!]((), x) = \{\!| x = 1 |\!\} \quad [\![\;]\!](x, y) = \{\!| y = k \cdot x |\!\} \quad [\![\;]\!]((), x) = \frac{1}{2}x^2$$

The interpretation of any Σ*-term is, therefore, the vertical(tensor product) and horizontal(categorical composition) combination of the interpretation of each generator that composes it.*

Thus, **GQA** is an extension of the previous algebra for affine relations **GAA**.

The extension is done through the addition of a new generator, the gray arrow generator, and its associated axioms. This is of particular interest, because is the interpretation of such generator that is lifting the category to the level of weighted relations instead of binary relations.

To understand its role, recall that every generator of **GAA** is interpreted as the indicator function of some affine subspace. In this picture, one may think of numbers flowing along the wires, and each diagram as encoding membership in some affine set. The gray arrow generator behaves differently, rather than enforcing a constraint on the values running through the wire, it *measures* them and produces a real-valued output. No number is transformed inside it; instead, a cost is emitted.

A useful analogy comes from Electronics. An electric circuit diagram is a formal diagram that allows you to represent an electric circuit with wires and devices attached to those wires. In general, what you see is current running in the wires and some components along it that alter the current. However, one may attach a measurement device, like an ammeter, in the circuit. That does not change the current, but it measures it, and outputs a number to outside the diagram.

Distinctively, the syntax admits an alternative probabilistic interpretation. Under this reading, the gray arrow generator encodes the information of a Gaussian distribution with mean 0, its semmantics being that of the function $(\bullet, x) \mapsto \mathcal{N}(0, x)$; categorically this encodes the notion of a state in Markov Categories [SS23; CJ19; Fri20]. Hence, **GQA** expresses linear functions of random variables, more precisely of normally distributed random variables, algebraically, turning composition into the act of probabilistic conditioning.

This interpretation of the syntax alludes to the understanding of the new added generator as the act of observing a particular random variable, thus getting a number out of it. Although such an extension is not explored in this work, we encourage the interested reader to pursue this direction [Ste+24; SS21]. We believe there exists a very strong interconnection between the algebraic semmantics of optimization and that of probabilistic programming [SS24], the notion of measuring in the convex setting being equivalent to that of observing the value of a random variable in the stochastic one, and the composition playing the role of variable elimination in the first and of conditioning in the latter. This lacks formality however, and we leave it to future works.

This provides a very meaningful intuition on how to interpret the new generator. Rather then transforming the values carrieds by wires, the grey arrow generator measures them and emits a real-valued cost. In this case, the semantics of the diagram is $(x, \bullet) \mapsto \frac{1}{2}\|x\|^2$, it measures the l2 norm of the number in the wires. This observation is central, since the idea of a generator that "measures" or to "outputs" values carried by wires is the conceptional foundation in our theory of Linear Programming develop in Section 6.

Taking all the Σ-terms and quotient by their axioms E we generate the free PROP of quadratic diagrams.

Definition 20 (The Free category of quadratic diagrams)**.** *We call* $FreeP_{\mathbf{GQA}}$

the category of string diagrams of $\boldsymbol{GQA}$

Observe that we have associated to each generator of Σ a visual diagram. This can be made precise through the notion of a *strict monoidal functor*. Concretely, the Joyal–Street coherence theorem [JS91] provides a canonical isomorphism of categories

$$\mathbf{FreeP}_{\mathbf{GQA}} \xrightarrow{\sim} \mathbf{Diag}(\Sigma),$$

where $\mathbf{Diag}(\Sigma)$ is the category of isotopy classes of string diagrams generated by Σ. Under this isomorphism, each generating morphism $g \in \Sigma$ is sent to the box labelled g with its input and output wires, composition $\circ$ is sent to vertical stacking of diagrams, and the tensor product $\otimes$ is sent to horizontal juxtaposition. The fact that this assignment is a *functor* is precisely what guarantees that diagrammatic reasoning is sound: any equation derivable from the axioms of **GQA** holds if and only if the corresponding diagrams are related by isotopy.

With the syntax in place, we now construct the corresponding semantic category. As established in the preceding section, the objects of interest are non-negative partial quadratic functions, or equivalently, quadratic optimization problems with equality constraints whose objective is bounded below by zero. These are encoded as *quadratic relations* where $x \in \mathbb{R}^n$ is related to $y \in \mathbb{R}^m$ through a partial non-negative quadratic function

$$f : \mathbb{R}^n \times \mathbb{R}^m \to [0, +\infty],$$

where $f(x, y)$ is interpreted as the quadratic cost associated to the pair (x, y), with $f(x, y) = +\infty$ indicating infeasibility. The following definition constructs the categorical equivalent of such notion.

Definition 21 (**QuadRel**)**.** *We define* **QuadRel** *as follows.*

Objects. *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the Euclidean space $\mathbb{R}^n$.*

Morphisms. *A morphism*

$$f : n \to m$$

is a partial non-negative quadratic function

Composition. *Given*

$$f : n \to m, \qquad g : m \to k,$$

their composite

$$g \circ f : n \to k$$

is defined by partial minimisation:

$$(g \circ f)(x, z) = \inf_{y \in \mathbb{R}^m} \left(f(x, y) + g(y, z) \right).$$

Identities. *For each n, the identity morphism*

$$\mathrm{id}_n : n \to n$$

is given by the quadratic form

$$\mathrm{id}_n(x, y) = \begin{cases} 0 & \text{if } x = y, \\ +\infty & \text{otherwise.} \end{cases}$$

Monoidal structure. *The tensor product on objects is addition:*

$$n \otimes m := n + m.$$

For morphisms

$$f : n_1 \to m_1, \qquad g : n_2 \to m_2,$$

their tensor product

$$f \otimes g : n_1 + n_2 \to m_1 + m_2$$

is defined by

$$(f \otimes g)(x_1, x_2, y_1, y_2) = f(x_1, y_1) + g(x_2, y_2).$$

From Proposition 4, composition is well defined. With these operations, **QuadRel** *forms a symmetric monoidal category (indeed a PROP).*

To prove that **GQA** presents **QuadRel**, we require several preliminary results. A central ingredient in completeness proofs of this kind is the existence of a *normal form*: a canonical diagrammatic representation to which every morphism in the category can be reduced by applying the axioms of the algebra.

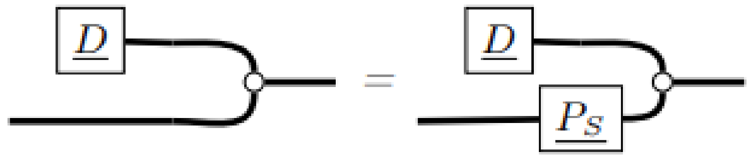
The structure of the normal form is typically guided by the key computational primitive of the semantic category in question. For example, in **GLA**, every linear relation can be represented by a pair of matrices A, B such that

$$(x, y) \in R \iff Ax = By \iff \begin{bmatrix} A & -B \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \iff \widetilde{A}\,\tilde{x} = 0,$$

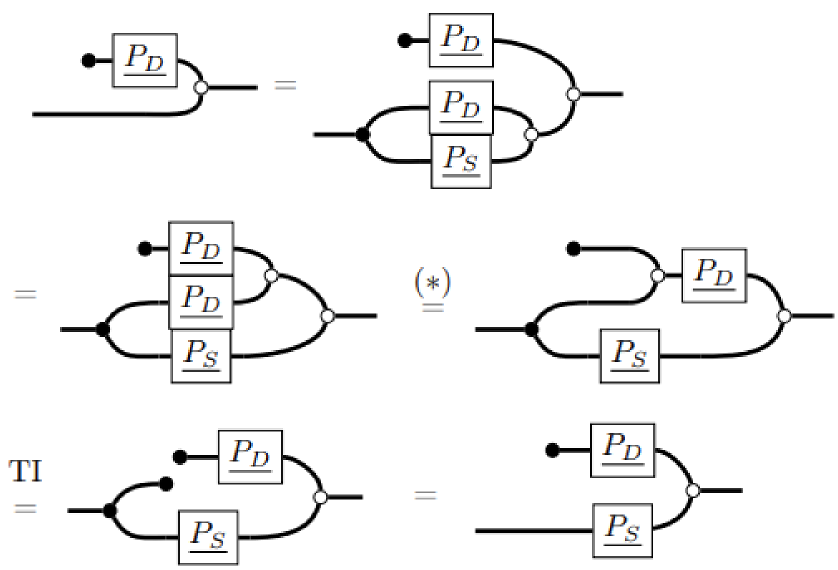
and the reduction to normal form proceeds through a diagrammatic argument that mirrors Gaussian elimination on the underlying linear system.

In the quadratic setting, the analogous role is played by the QR factorization. Just as Gaussian elimination reduces a linear relation to a canonical matrix kernel form, QR factorization reduces a quadratic diagram to a canonical form by orthogonally decomposing the underlying matrix. This is the appropriate primitive because the objective function $\|y\|^2$ is invariant under orthogonal transformations, so the axioms of **GQA** can simulate QR steps diagrammatically. The following results make this precise.

Proposition 5. *If S, D are complementary subspaces, then we can prove that:*



Proof. From the fact that $P_S + P_D = I$, such that P_S, P_D are the projection matrices over subspaces S and D, we show:

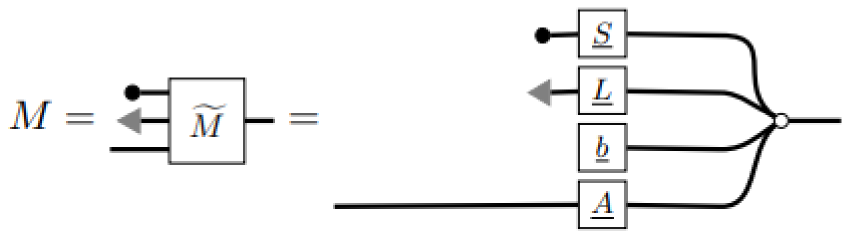


$\square$

Above result is an analogous way of saying that: $\{(x, y) | \exists z \in D, z + x = y\} = \{(x, y) | \exists z \in D, z = y - x\}$, by the orthogonal unique decomposition theorem $\{(x, y) | \exists z \in D, z = y - x_S - x_D\} = \{(x, y) | \exists z \in D, z + x_D = y - x_S\}$ Since $x_D, z \in D \implies x_D + z = \tilde{z} \in D$ and, by the uniqueness of the decomposition, $y = \tilde{z} + x_S \implies \tilde{z} = y_D, y_S = x_S$. $\{(x, y) | \exists \tilde{z} \in D, \tilde{z} = y - x_S\}$. Visually, one can interpret above diagram as the idea that summing a dangling wire with a subspace D is equivalent to ask for a pair $(y_{D^\perp}, y)$.

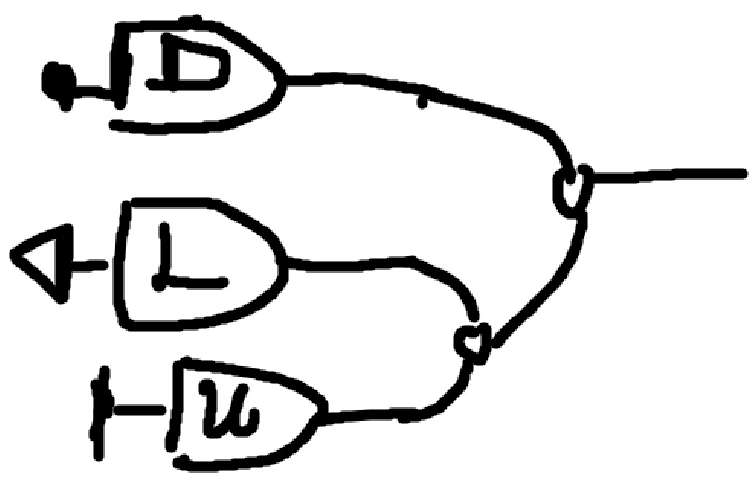
The following remmark establishes a preliminary normal form for the directed fragment $\overrightarrow{\mathbf{GQA}}$, which will serve as the key stepping stone toward the full normal form theorem. Intuitively, it says that any diagram in this fragment can be reorganised into a canonical block structure determined by a subspace S and compatible matrix data.

Remark 6. *Let M be any diagram generated by the fragment $\overrightarrow{\mathbf{GQA}}$ (recall Definition 12) together with the generator* $\bullet\!-$. *By collecting terms, we can write it as a diagram of the form*



such that $S \subseteq \mathbb{R}^n$ is a vector subspace, $b \in S^\perp$, $\mathrm{im}(LL^\top) \subseteq S^\perp$, and $\mathrm{im}(A) \subseteq S^\perp$ (By Proposition 5).

Proposition 6. *Let N be a diagram in the followinf form*



with data (D, μ, L), where $D \subseteq \mathbb{R}^n$ is a vector subspace, $\mu \in D^\perp$, L is a lower-triangular matrix with and $\mathrm{im}(LL^\top) \subseteq D^\perp$. Its semantic interpretation in **QuadRel** *is the partial non-negative quadratic function*

$$[N](x) = \frac{1}{2}(x-\mu)^\top LL^\top(x-\mu) + \{|D^\perp + \mu|\}(x),$$

Moreover, two canonical block-form diagrams N_1, N_2 with data (D_1, μ_1, L_1) and (D_2, μ_2, L_2) satisfy $[N_1] = [N_2]$ in **QuadRel** *if and only if*

$$D_1 = D_2, \qquad \mu_1 = \mu_2, \qquad L_1 L_1^\top = L_2 L_2^\top.$$

Proof. The first claim follows by direct computation of the interpretation functor $[-]$ applied to the block form. The L-block contributes the quadratic term $\frac{1}{2}(x-\mu)^{\top}LL^{\top}(x-\mu)$, the subspace D contributes the indicator $\iota_{D^{\perp}}$, and the vector μ shifts the centre of the feasible region.

For the equivalence, the $(\Leftarrow)$ direction is immediate from the first claim. For $(\Rightarrow)$, suppose $[N_1] = [N_2]$ pointwise on $\mathbb{R}^n$.

The domain of finiteness of $[N_i]$ is the affine subspace $D_i^{\perp} + \mu_i$. Since $[N_1] = [N_2]$, these domains coincide: $D_1^{\perp} + \mu_1 = D_2^{\perp} + \mu_2$. Two cosets are equal only if their direction subspaces agree, so $D_1^{\perp} = D_2^{\perp}$, hence $D_1 = D_2$. Write $D := D_1 = D_2$.

The coset equality now gives $\mu_1 - \mu_2 \in D^{\perp}$. Evaluating both functions at $x = \mu_1$:

$$0 = [N_1](\mu_1) = [N_2](\mu_1) = \frac{1}{2}(\mu_1 - \mu_2)^{\top} L_2 L_2^{\top} (\mu_1 - \mu_2).$$

Since $L_2L_2^{\top}$ is positive semidefinite and its restriction to $D^{\perp}$ is positive definite (as $\mathrm{im}(L_2L_2^{\top}) \subseteq D^{\perp}$ with L_2 of full rank on $D^{\perp}$), the equation above forces $\mu_1 - \mu_2 = 0$, i.e. $\mu_1 = \mu_2$.

With $D_1 = D_2$ and $\mu_1 = \mu_2 =: \mu$ established, the identity $[N_1](x) = [N_2](x)$ for all $x \in D^{\perp} + \mu$ yields

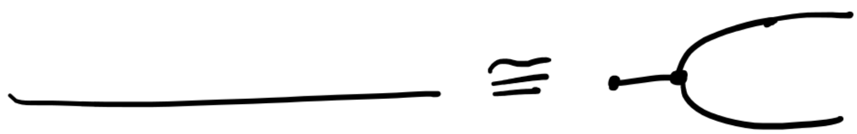
$$(x-\mu)^{\top} L_1 L_1^{\top} (x-\mu) = (x-\mu)^{\top} L_2 L_2^{\top} (x-\mu) \qquad \forall\, x \in D^{\perp} + \mu.$$

Setting $v = x - \mu \in D^{\perp}$, this is a quadratic identity on $D^{\perp}$. By polarisation it implies $L_1L_1^{\top} = L_2L_2^{\top}$ as operators on $D^{\perp}$, and since both sides vanish on D, equality holds on all of $\mathbb{R}^n$. $\square$

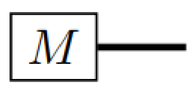
We now state and prove the main normal form theorem for **GQA**. The theorem states that every diagram can be decomposed into a purely relational part — a morphism in **GQA** — and a non-negative scalar, corresponding respectively to the constraint structure and the quadratic cost of the underlying optimization problem. The proof proceeds by a diagrammatic version of QR factorization: orthogonal transformations are simulated by axiom applications, progressively eliminating generators until the canonical form is reached.

Theorem 2 (**GQA** Normal Form)**.** *Every diagram $M \in \overrightarrow{\mathbf{GQA}}$ can be rewritten, modulo the axioms E, as $M' \otimes c$ where $M' \in \mathbf{GQA}$ and $c \in [0, +\infty]$ is a scalar of the form:*

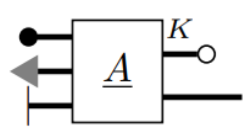
Proof. In a hypergraph category, there is a bijection between $\mathrm{Hom}(c, d)$ and $\mathrm{Hom}(\mathbb{1}, c \otimes d)$, where $\mathbb{1}$ is the monoidal unit. Diagrammatically, this corresponds to bending wires from the left side to the right using the cap and cup morphisms of the compact closed structure:



Morphisms from the monoidal unit are called *states*. This means that any diagram can be reorganised so that all dangling wires appear on one side. Collecting all dangling wires to the left and all generators that are morphisms from the unit object to the right, we obtain a diagram of the form:

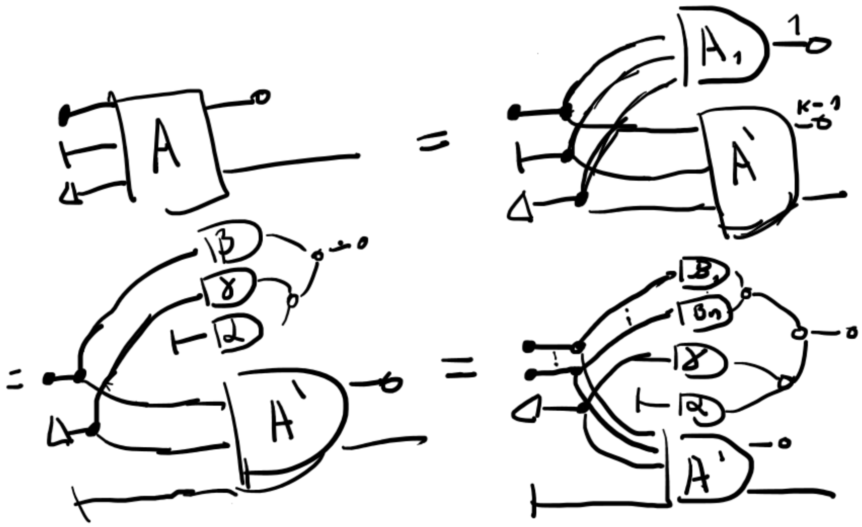


where M is a diagram of arity 0. We now separate the $\multimap$ generators from the remaining ones, and collect the dangling wires accordingly. Without loss of generality, assume there are K such generators:

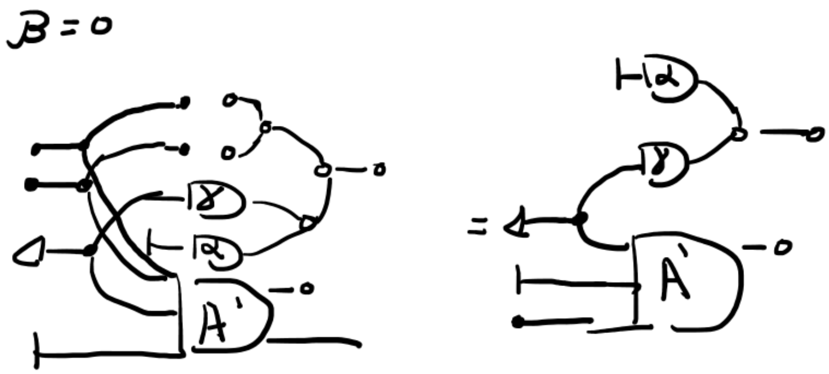


Here A is a fragment of $\overrightarrow{\mathbf{GLA}}$ and therefore represents a matrix [BSZ17]. To realize this, just recall that we can take the co-sum and co-copy and transform them into the copy and the sum by bending the wires with a spider.

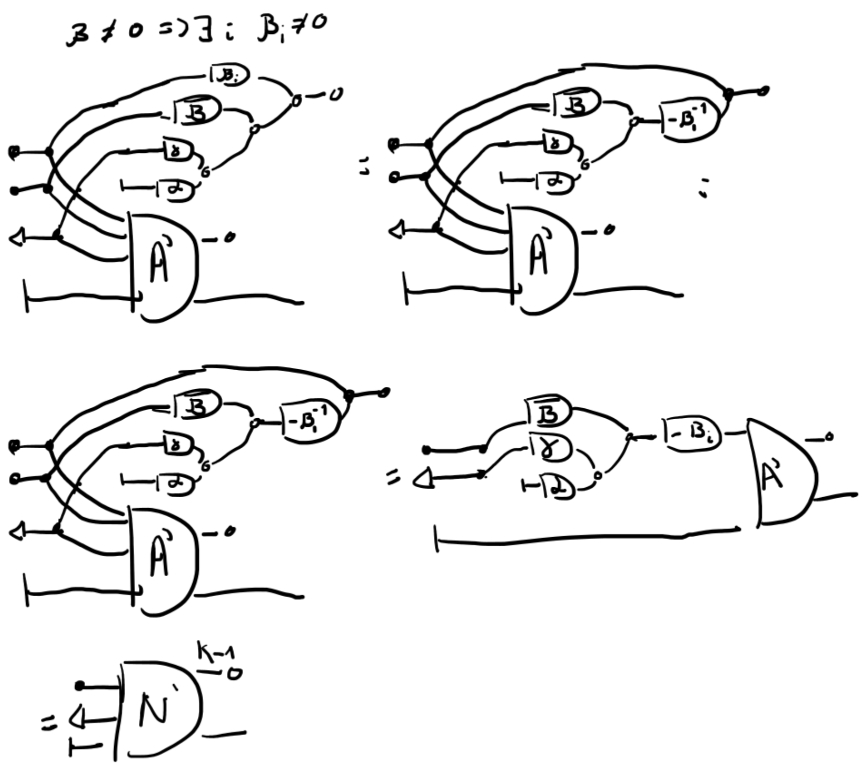
Applying the axioms, we derive:



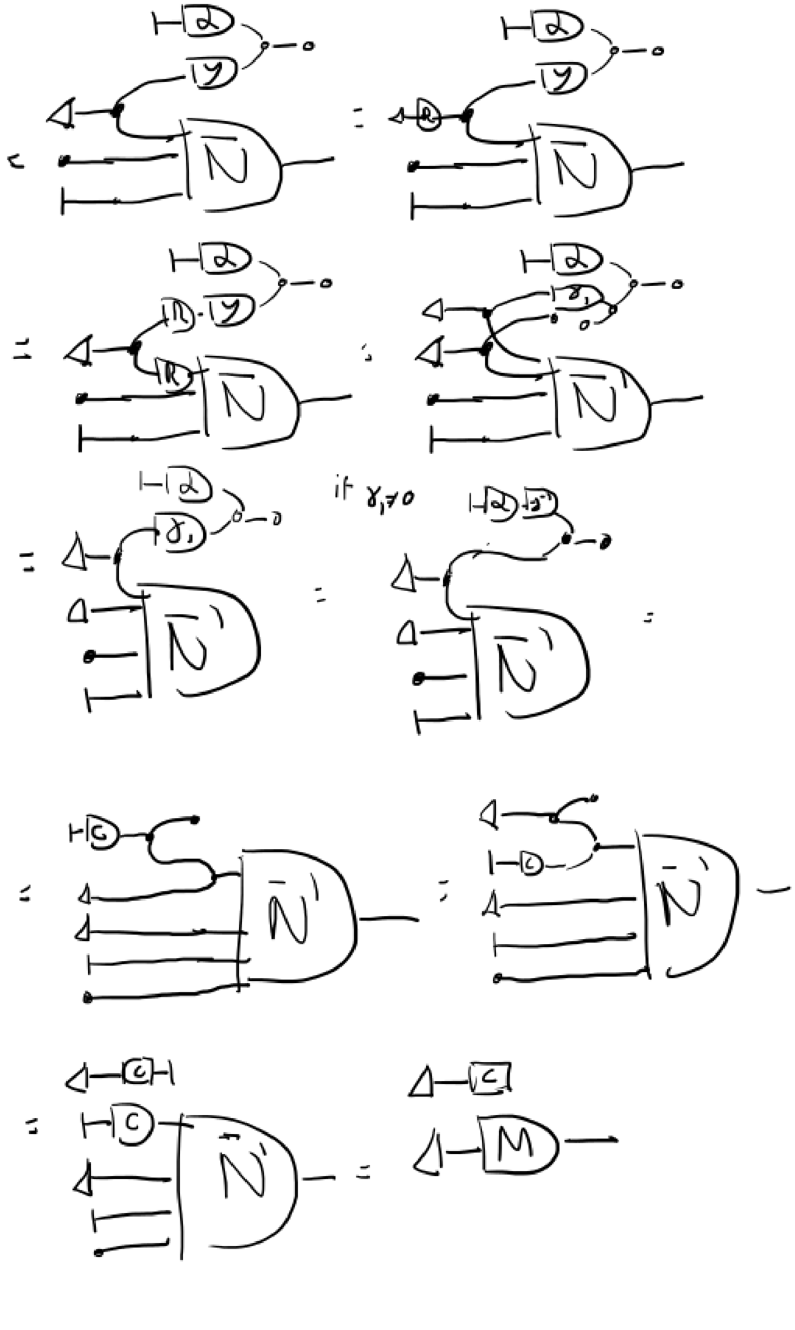
We use the fact that a matrix can be decomposed into rows. We pick A_1 as the row satisfying $A_1 x = 0$ and decompose it into scalar components α, β, γ — one-dimensional row vectors — such that $\alpha x_1 + \beta x_2 + \gamma x_3 = 0$. If $\beta = 0$, the corresponding black dot generator is eliminated immediately:



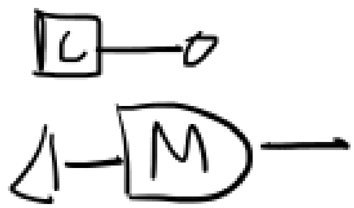
If $\beta \neq 0$, then there exists an index i such that $\beta_i \neq 0$, and we can eliminate one occurrence of the white dot and one of the black dot simultaneously:



By induction on the number of rows, we repeat this process for all rows with $\beta \neq 0$, leaving only rows with $\beta = 0$. The diagram reduces to:



Here R is an orthogonal matrix such that $R\gamma = (\gamma_1, 0, \dots, 0)$. If $\gamma_1 \neq 0$, the normal form is achieved. If $\gamma_1 = 0$, the wire disconnects trivially and we are left with:



In both cases the diagram has been reduced to the claimed normal form $M' \otimes c$, completing the proof. $\square$

With the normal form established, we now have all the ingredients needed to prove the three properties — soundness, expressivity, and completeness — that together guarantee that **GQA** is a faithful and complete axiomatisation of **QuadRel**. We treat each in turn.

Lemma 1 (Soundness)**.** *If* $s = t$ *in* $\mathrm{FreeP}_{\mathbf{GQA}}$*, then* $[s] = [t]$ *in* **QuadRel**.

Proof. Since **AffRel** embeds into **QuadRel** via the indicator function, all axioms inherited from **GAA** are already sound in **QuadRel**. It therefore suffices to verify soundness for the newly introduced axioms involving the gray arrow generator, which is done by direct computation of the corresponding semantic interpretations:

$$
\begin{aligned}
[\![\blacktriangleleft\!-\!\boxed{R}\!-]\!]\,(\mathbf{y}) &= \inf_{\mathbf{x}} \left\{ \frac{1}{2}||\mathbf{x}||^2 + \{\!|\, \mathbf{y} = R\mathbf{x} \,|\!\} \right\} \\
&= \frac{1}{2}||R^{-1}\mathbf{y}||^2 = \frac{1}{2}||\mathbf{y}||^2 \\
&= [\![\blacktriangleleft\!\!-\!\!-]\!]\,(\mathbf{y}) \\
[\![\blacktriangleleft\!\!-\!\!-\!\bullet]\!] &= \inf_{x} \left\{ \frac{1}{2}x^2 + 0 \right\} = 0 = [\![\;\;]\!] \\
[\![\blacktriangleleft\!\!-\!\!-\!\circ]\!] &= \inf_{x} \left\{ \frac{1}{2}x^2 + \{\!|\, x = 0 \,|\!\} \right\} = \frac{1}{2} \cdot 0^2 = [\![\;\;]\!]
\end{aligned}
$$

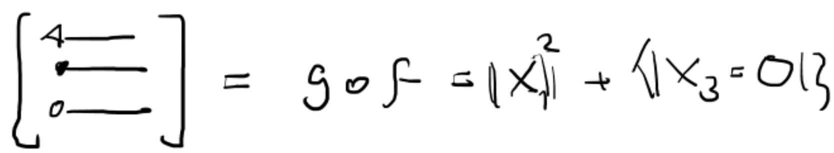
$\square$

Lemma 2 (Expressivity)**.** *For every $f : m \to n$ in* **QuadRel***, there exists a Σ-term $s : m \to n$, equivalently a morphism $s \in \mathrm{Hom}_{\mathrm{FreeP}_{\mathbf{GQA}}}(m, n)$, such that $[s] = f$.*

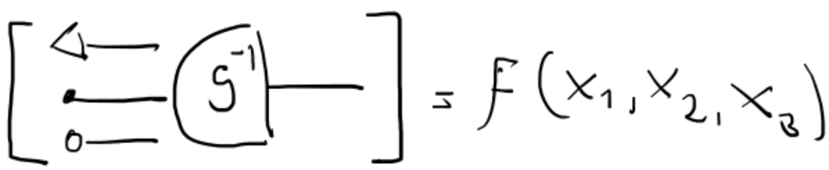
Proof. By Proposition 3, for every non-negative partial quadratic function f there exists an invertible coordinate change $\mathbf{g}$ such that

$$\mathrm{g}(f(x_1, x_2, x_3)) = \frac{1}{2}\|x_1\|^2 + \mathbf{1}_{\{0\}}(x_3),$$

which is represented diagrammatically as:



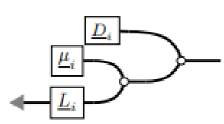
Since the coordinate change $\mathbf{g}$ is itself expressible as a morphism in **GQA**, the general case follows by composition:



$\square$

Lemma 3 (Completeness)**.** *If $[s] = [t]$ in* **QuadRel***, then $s = t$ in $\mathrm{FreeP}_{\mathbf{GQA}}$.*

Proof. We reduce both s and t to their respective normal forms $s = M_1 \otimes c_1$ and $t = M_2 \otimes c_2$ using the Normal Form Theorem. Since M_1, M_2 lie in the fragment $\overrightarrow{\mathbf{GQA}} \cup \{\bullet\!-\}$, Lemma 6 allows us to write them in the canonical block form:



where D_i is a vector subspace, $\mu_i \in D_i^{\perp}$, L_i is lower triangular, and $\mathrm{im}(L_i L_i^{\top}) \subseteq D_i^{\perp}$. By Proposition 6, $[N_1] = [N_2]$ if and only if $D_1 = D_2$, $\mu_1 = \mu_2$, and $L_1 L_1^{\top} = L_2 L_2^{\top}$.

If $AA^{\top} = BB^{\top}$, then there exists an orthogonal matrix R such that $A = BR$, and the axioms of **GQA** allow us to rewrite one diagram into the other by applying the corresponding orthogonal transformation diagrammatically. Appealing to the completeness of **GAA** for the affine fragment, we conclude $N_1 = N_2$ in $\mathrm{FreeP}_{\mathbf{GQA}}$. Since $M_i = N_i \otimes c_i$ and $c_1 = c_2$ follows from $[s] = [t]$, we obtain $M_1 = M_2$, as desired. □

Combining the three lemmas, the main presentation theorem follows immediately.

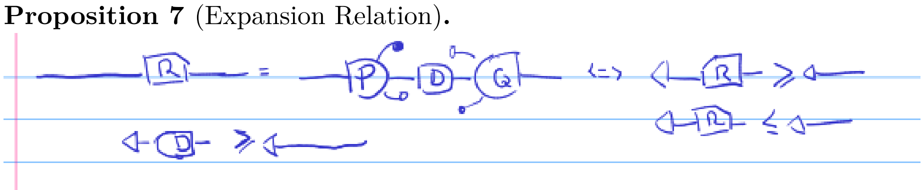
Theorem 3. **QuadRel** *is presented by* $\mathrm{FreeP}_{\mathbf{GQA}}$.

Proof. Soundness, expressivity, and completeness were established by the preceding three lemmas. Together they imply that the canonical PROP morphism $\mathrm{FreeP}_{\mathbf{GQA}} \to \mathbf{QuadRel}$ is full, faithful, and essentially surjective, hence an isomorphism of PROPs. □

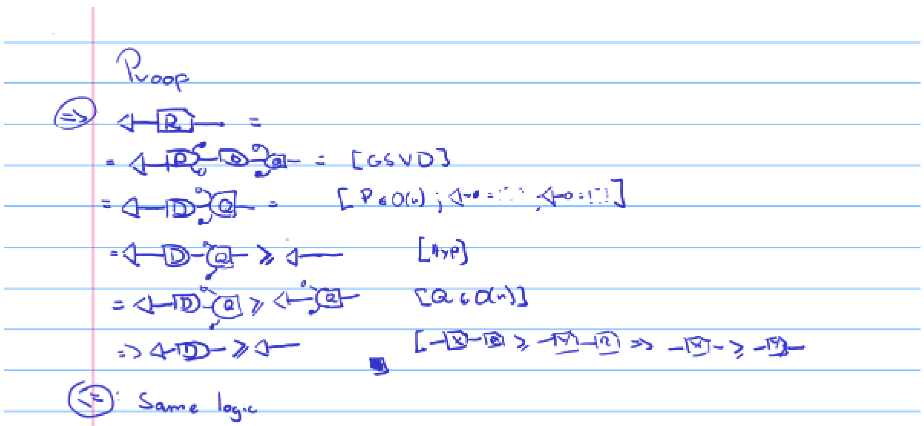
4.3 Results in GQA

Having established that **GQA** completely axiomatises **QuadRel**, we now demonstrate the expressive power of the calculus by deriving several non-trivial results purely through string diagram manipulation. These propositions illustrate that the diagrammatic language is not merely a notational device, but a genuine proof calculus in which geometric and algebraic arguments about quadratic optimization can be carried out compositionally. We begin with two fundamental structural results — the expansion and contraction relations — before moving to orthogonal projections and the generalized singular value decomposition.

Proposition 7 (Expansion Relation)**.**

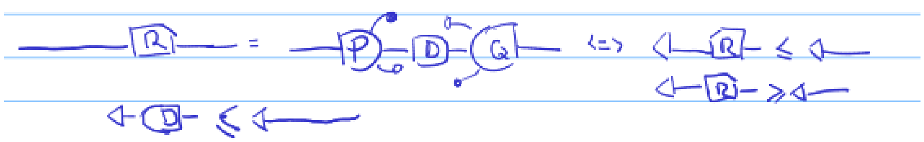


Proof.



□

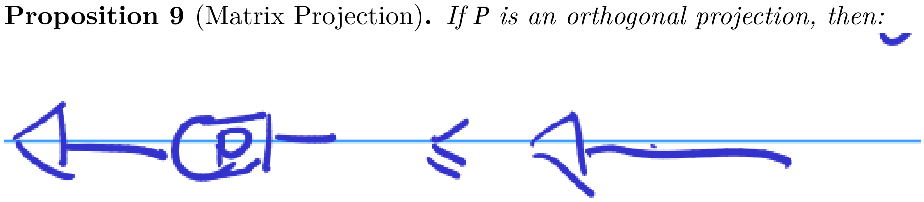
Proposition 8 (Contraction Relation)**.**



Proof. The proof follows by an entirely analogous diagrammatic argument to that of the Expansion Relation above, applying the axioms of **GQA** in the reverse direction. □

The following proposition establishes the diagrammatic representation of orthogonal projections. This is a key result, as projections appear naturally in least-squares problems whenever the system $Ax = b$ is inconsistent, and the pseudoinverse computation relies essentially on projecting onto the image of A.

Proposition 9 (Matrix Projection)**.** *If P is an orthogonal projection, then:*

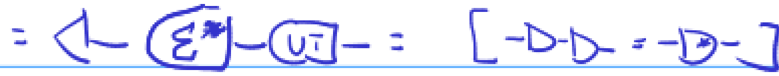


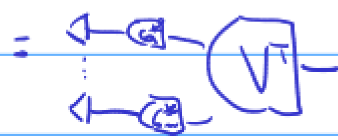
Proof.

 [S.V.D]

 $[V V^T = Id]$

 $[U \in O(n)]$

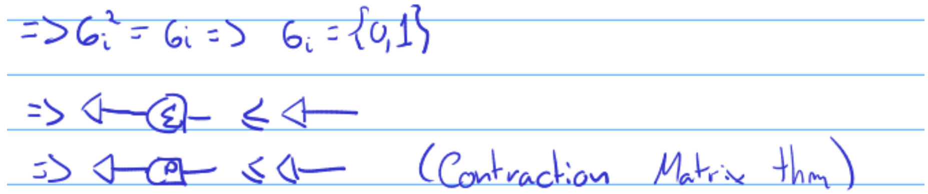


 $[\Sigma^* \text{ is diag}]$

 $[\Sigma^* = \Sigma\Sigma = \Sigma^2]$

 $[\Sigma^* = \Sigma]$

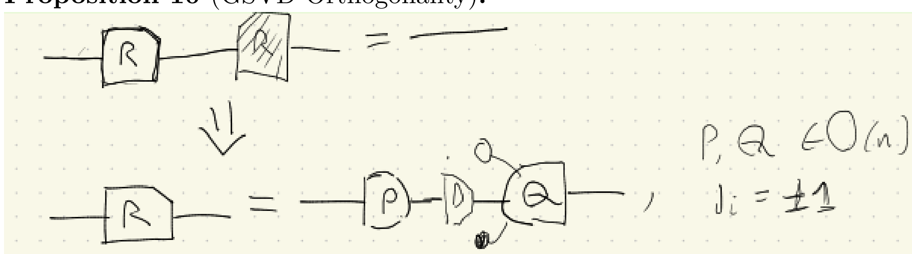
$$\Rightarrow \sigma_i^2 = \sigma_i \Rightarrow \sigma_i = \{0,1\}$$



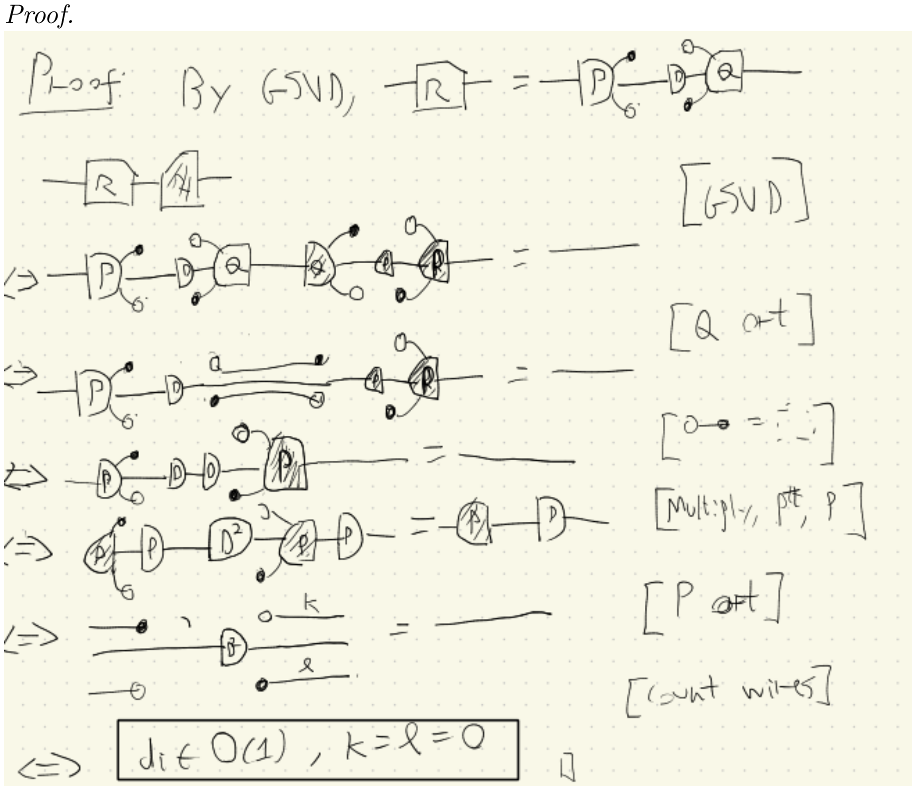
∎

The next result extends the orthogonality argument to the setting of generalized singular value decompositions.

Proposition 10 (GSVD Orthogonality)**.**



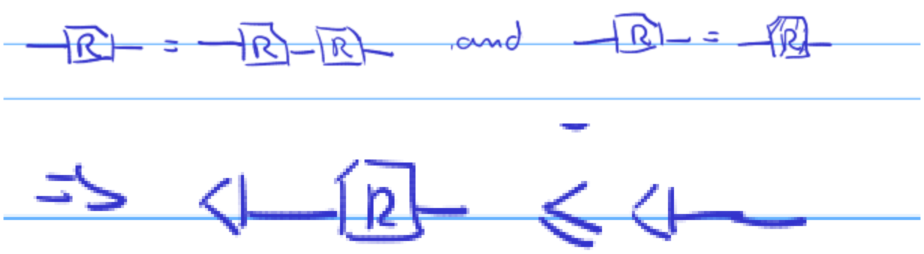
Proof.



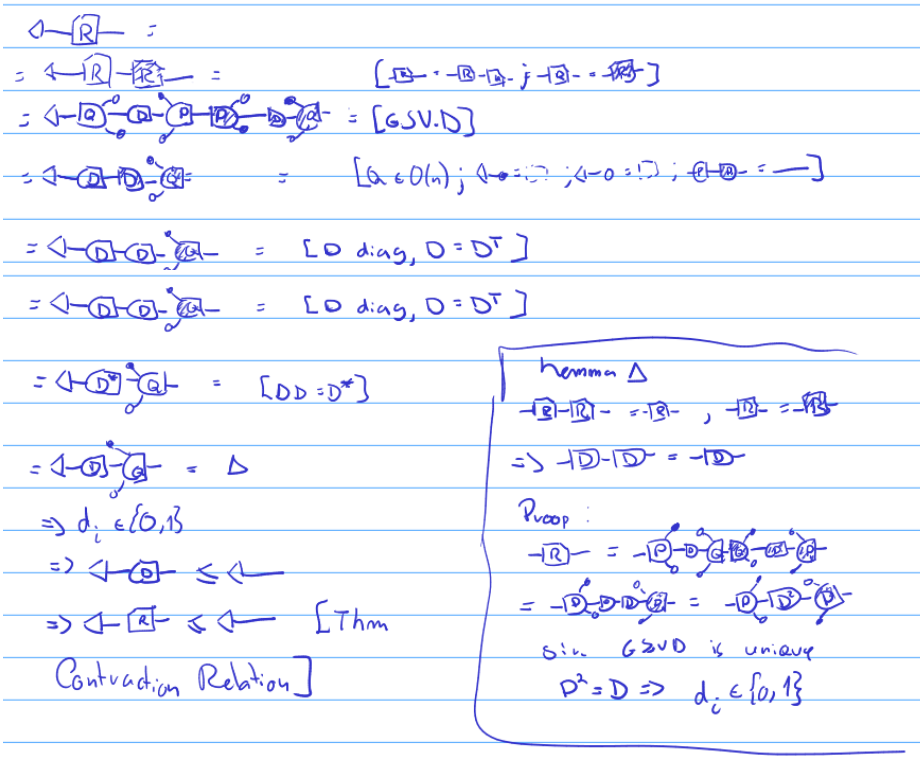
□

Finally, the Relation Projection proposition combines the preceding results to characterise how orthogonal projections interact with quadratic relations.

Proposition 11 (Relation Projection)**.**



Proof.



□

Proposition 12 (The OLS estimator)**.**

5 Polyhedral Relations

While the previous section treated quadratic costs, we now turn to a combinatorially richer class of objects: polyhedra and polyhedral cones. These are the fundamental geometric structures underlying linear programming, and they admit a categorical treatment entirely analogous to the one developed for quadratic relations. The key difference is that composition here is governed by Fourier-Motzkin Elimination (FME) rather than QR factorization: projecting a polyhedron onto a subspace is precisely variable elimination, and it is this operation that categorical composition will simulate diagrammatically.

Unlike the previous section, the theory developed here remains a within the category of binary relations since morphisms are still indicator functions of subsets of $\mathbb{R}^n \times \mathbb{R}^m$, encoding feasibility constraints rather than objective functions. In this sense, **GPA** does not yet enter the weighted relational world of not being able to express more meaningful optimization problems. This is nonetheless a critical step in the development. Axiomatizing linear inequality constraints diagrammatically, **GPA** provides the Symmetric Monoidal Inequality Theory (SMIT) [BDS21] that will serve as the structural foundation for our final contribution, **GLP**, where these constraints are combined with a linear objective to form full parametric linear programs.

We begin by recalling the standard definitions of polyhedral cones and polyhedra, before introducing the two graphical algebras of $\mathbf{GPA}_{\vdash}$ [BDS21] for polyhedral cones and **GPA** for general polyhedra, proving that each completely axiomatises its corresponding semantic category.

Definition 22. *A set* $C \subseteq \mathbb{R}^n$ *is called a* polyhedral cone *if there exists a matrix* $A \in \mathbb{R}^{m\times n}$ *such that*

$$C = \{x \in \mathbb{R}^n \mid Ax \geq 0\}.$$

Definition 23. *A set* $P \subseteq \mathbb{R}^n$ *is called a* polyhedron *if there exist* $A \in \mathbb{R}^{m\times n}$ *and* $b \in \mathbb{R}^m$ *such that*

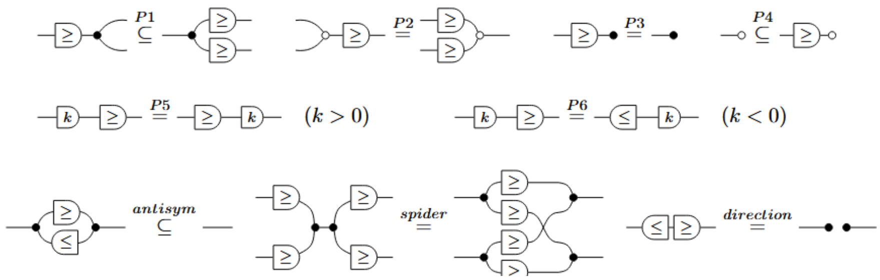
$$P = \{x \in \mathbb{R}^n \mid Ax \geq b\}.$$

Note that a polyhedral cone is a special case of a polyhedron with $b = 0$. The distinction is algebraically significant: cones are closed under non-negative scaling and admit a purely homogeneous representation, while general polyhedra may be translated. This distinction is reflected in the two graphical algebras we now define, where $\mathbf{GPA}_{\vdash}$ handles the homogeneous case and **GPA** extends it with an affine generator.

Definition 24 (Graphical Polyhedral Algebra)**.** *We call* **GPA** *the SMT generated by extending* **GAA** *with the following generator and its semantics:*

$$[\![-\boxed{\geq}-]\!] = \{ (x, y) \mid x, y \in \mathsf{k}, x \geq y \}.$$

along with the following additional equations:



Definition 25 (Graphical Polyhedral Cones Algebra)**.** *We call* $\mathbf{GPA}_{\vdash}$ *the SMT generated by* **GPA** *but excluding the affine generator* $\vdash$ *and its associated equations. This fragment axiomatises polyhedral cones, where no translation is present.*

Definition 26 (The Free Categories of Polyhedral Diagrams)**.** *We call* $\mathrm{FreeP}_{\mathbf{GPA}}$ *the PROP freely generated by* **GPA***, and* $\mathrm{FreeP}_{\mathbf{GPA}_{\vdash}}$ *the PROP freely generated by* $\mathbf{GPA}_{\vdash}$*.*

We now define the two semantic categories that these algebras will be shown to axiomatise. Just as **AffinRel** captured affine space as morphisms, the categories **PCRel** and **PolyRel** capture polyhedral cones and polyhedra as morphisms, with relational composition playing the role of variable elimination.

Definition 27 (The prop **PCRel** of polyhedral cones)**.** *Let k be an ordered field, in this case the real numbers. The prop* $\mathbf{PCRel}_k$ *is defined as follows:*

- *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space k^n.*
- *A morphism $n \to m$ is a polyhedral cone $R \subseteq k^n \times k^m$ for which there exist $p \in \mathbb{N}$ and $A \in k^{p \times (n+m)}$ such that*

$$R = \left\{ (x, y) \in k^n \times k^m \;\middle|\; A \begin{pmatrix} x \\ y \end{pmatrix} \geq 0 \right\}.$$

- *Composition is relational composition.*
- *The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is the Cartesian product.*

Definition 28 (The prop **PolyRel** of polyhedra)**.** *Let k be an ordered field. The prop* $\mathbf{PolyRel}_k$ *is defined as follows:*

- *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space k^n.*
- *A morphism $n \to m$ is a polyhedron $R \subseteq k^n \times k^m$ for which there exist $p \in \mathbb{N}$, $A \in k^{p\times(n+m)}$, and $b \in k^p$ such that*

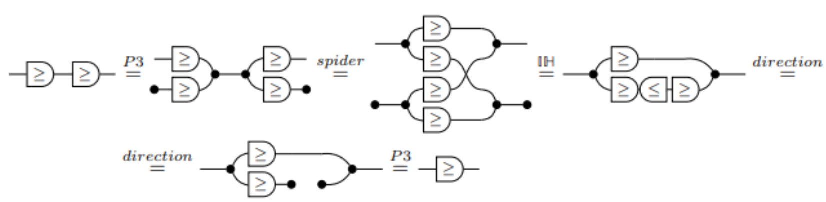
$$R = \left\{ (x, y) \in k^n \times k^m \;\middle|\; A \begin{pmatrix} x \\ y \end{pmatrix} + b \geq 0 \right\}.$$

- *Composition is relational composition.*
- *The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is the Cartesian product.*

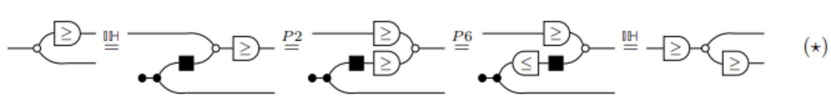
5.1 The algebra $\mathbf{GPA}_{\vdash}$ and PCRel

We now develop the presentation theory for the homogeneous case. The central computational primitive here is Fourier–Motzkin Elimination, which projects a polyhedral cone onto a lower-dimensional space by eliminating one variable at a time.

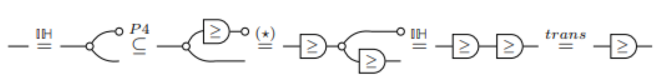
To accomplish it, we need some previous results regarding small diagrammatic operations. The first result we prove is the transitivity property of the generator, i.e. $- \geq - \geq - = - \geq -$



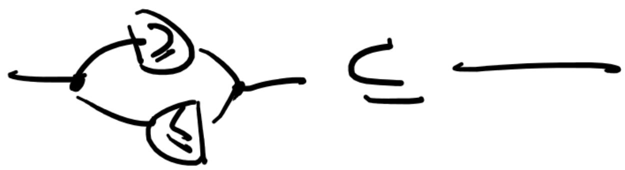
Next, we show that $\geq$ is reflexive

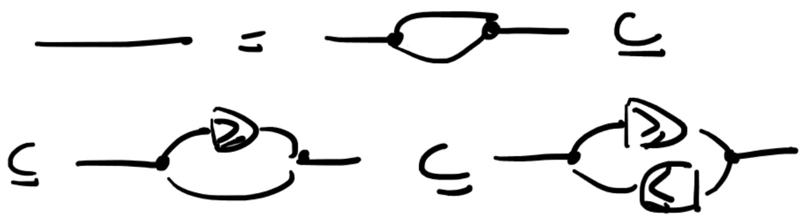


is used below to show that $\geq$ is reflexive:



and antisymetric

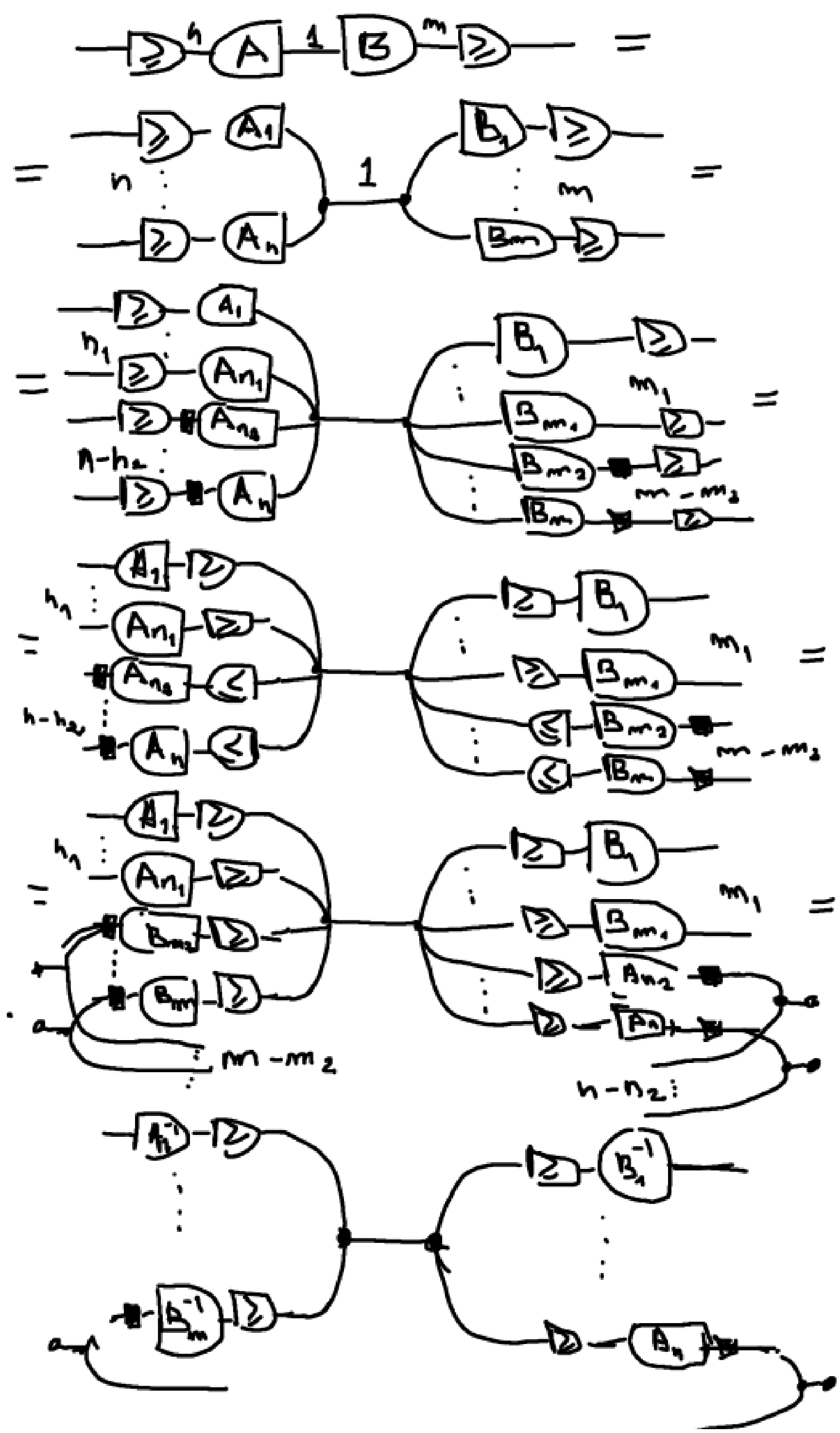


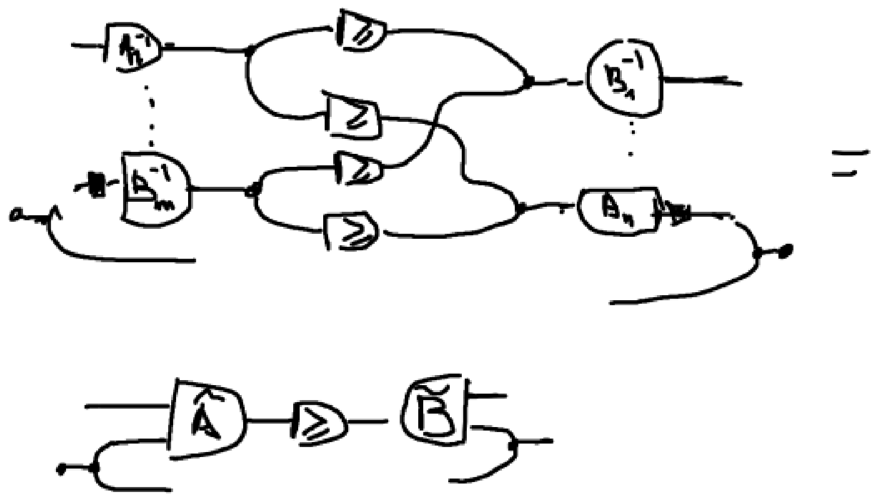


With those results in places we are ready to prove the diagrammatic FME:

Proposition 13 (Fourier-Motzkin Elimination)**.** *The following diagrammatic procedure is equivalent to perform the FME algorithm*

Proof. The proof proceeds by induction on the number of inequalities.

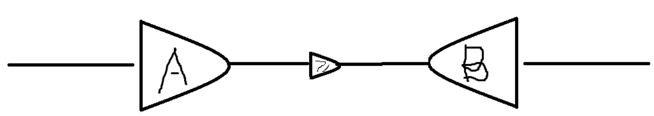




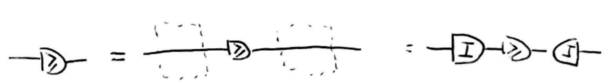
Now we show the inductive step for $n+1$: $\square$

With FME available as a diagrammatic operation, we can now prove the normal form theorem for $\mathbf{GPA}_{\vdash}$. We proceed by structural induction, handling generators, composition, and tensor product in turn, using FME steps to push all inequality generators into a canonical block form.

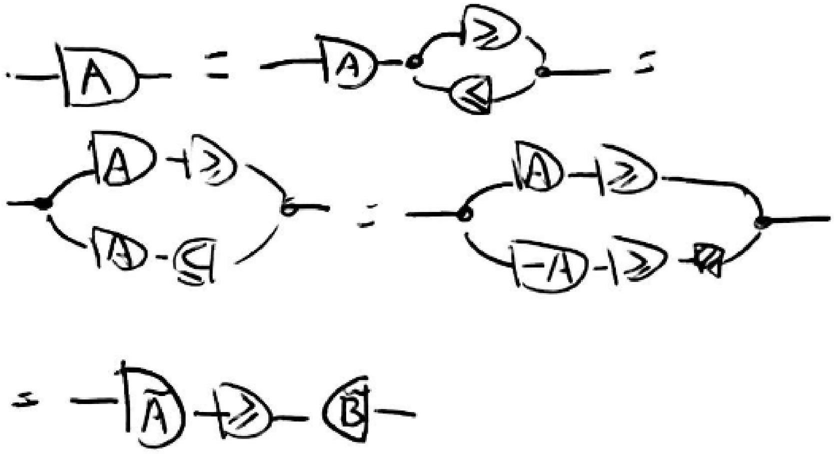
Theorem 4 (First Normal Form). *Every diagram in* $\mathbf{GPA}_{\vdash}$ *admits an isomorphic representation, modulo the axioms, of the following form:*



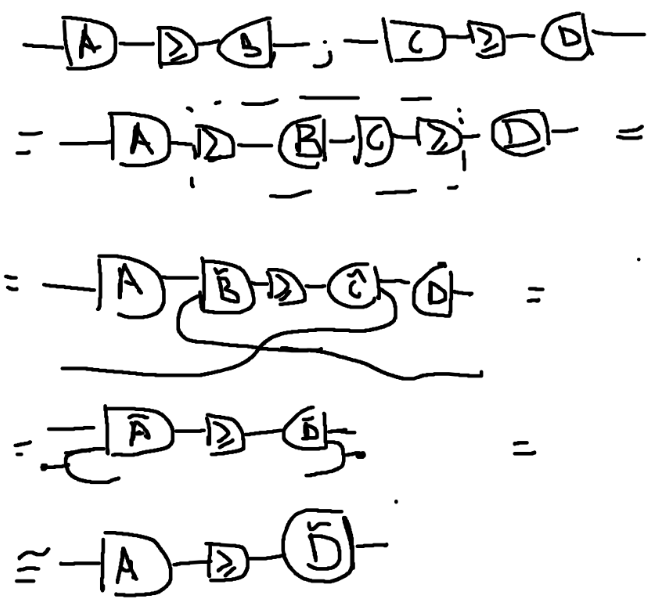
Proof. We proceed by structural induction. The two base cases are as follows. First, the $\geq$ generator:



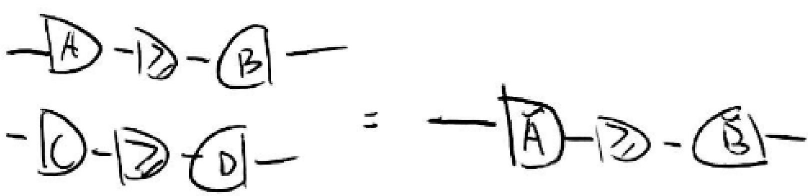
Second, for any diagram c in **GLA**:



For the inductive step, we consider composition of two diagrams $c;d$ in $\mathbf{GPA}_{\vdash}$. By the inductive hypothesis, both c and d are already in normal form:



And for the tensor product:



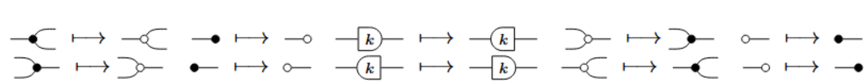
∎

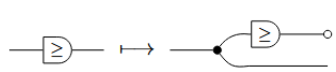
Working with a normal form defined up to isomorphism is in principle weaker than requiring syntactic equality of terms. This choice is motivated by the compact-closed structure, which provides a canonical bijection $\mathrm{Hom}(A, B) \cong \mathrm{Hom}(\mathbb{1}, A^* \otimes B)$, allowing input wires to be treated symmetrically with output wires but only up to this isomorphism, not up to equality. As shown in [BDS21], the same completeness argument can be carried out using strict diagrammatic equality; however, the derivations become significantly more involved. Since every result established for the isomorphic normal form carries over to the strict one, the only difference being the need to bend wires at each step, we adopt the more convenient convention throughout.

This idea was already encountered in Section 4.2, but it is worth pausing to appreciate its significance. The compact-closed structure does not merely allow a more convenient notation; it enables diagrammatic reasoning to serve as a rigorous replacement for coordinate-based algebraic manipulation while remaining extremely intuitive: bending a wire to the left or to the right does exactly what one would expect. Even more, a statement that would classically require several lines of matrix algebra, tracking indices, transposing, and substituting, can instead be verified by a sequence of local wire manipulations, each justified by a single axiom.

This first normal form expresses every diagram as a system of inequalities applied to a linear map. A dual normal form can be obtained by applying the polar operator, which we now define. The polar operator plays the role of duality in the polyhedral setting, exchanging the H-representation (intersection of halfspaces) with the V-representation (conic hull of generators), in direct analogy with the Farkas lemma [BDZ21].

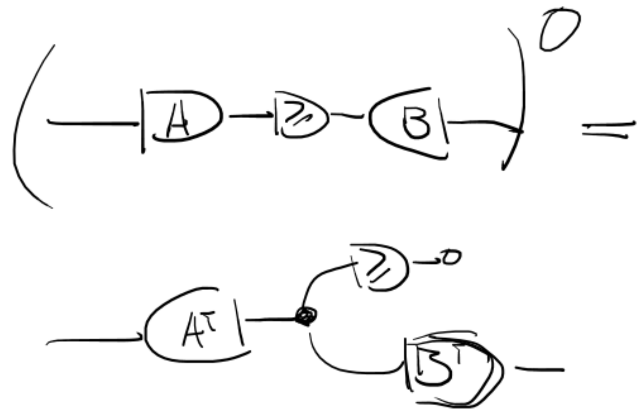
Definition 29 (Polar Operator). *The* polar operator $-^\circ$ *is the involution on generators defined by:*



 $\quad (c\,;d)^\circ = c^\circ\,;d^\circ \quad (c \oplus d)^\circ = c^\circ \oplus d^\circ$

Its action on diagrams is defined functorially by $(a \circ b)^\circ = a^\circ \circ b^\circ$.

Proposition 14 (Second Normal Form)**.** *Every diagram* $s \in \mathbf{GPA}_{\vdash}$ *can be written as:*

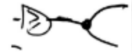


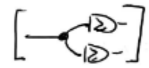
Proof. Apply the polar operator to the first normal form of every diagram $c \in \mathbf{GPA}_{\vdash}$. Since the polar operator is an involution and acts functorially, the result is again a valid diagram in $\mathbf{GPA}_{\vdash}$ in the claimed form. ☐

The second normal form expresses every diagram as a conic combination of column vectors, which is precisely the V-representation of a polyhedral cone. This is the representation that will drive the completeness proof: two diagrams that are semantically equal must have the same conic hull, and the axioms of $\mathbf{GPA}_{\vdash}$ are exactly sufficient to witness this equality diagrammatically. We now proceed to establish the three properties.

Lemma 4 (Soundness)**.** *If* $s = t$ *in* $\mathrm{FreeP}_{\mathbf{GPA}_{\vdash}}$*, then* $[s] = [t]$ *in* $\mathbf{PCRel}$.

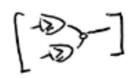
Proof. All axioms except those involving the newly introduced $\geq$ generator are already present in **GAA** and therefore verified. It suffices to check that the new axioms preserve the semantics:

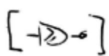
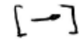
 $= \{(x, \begin{pmatrix}y_1\\y_2\end{pmatrix}) \mid x \geq y_1 \wedge x \geq y_2 \wedge y_1 = y_2\}$

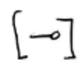
$\subseteq \{x, \begin{pmatrix}y_1\\y_2\end{pmatrix} \mid x \geq y_1 \wedge x \geq y_2\} =$ 

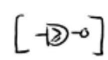
 $= \{\begin{pmatrix}x_1\\x_2\end{pmatrix}, y \mid x_1 + x_2 \geq y\} =$

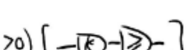
$\{(\begin{pmatrix}x_1\\x_2\end{pmatrix}, y) \mid \exists c,d\ x_1 + x_2 \geq c + d \wedge c + d = y\} =$

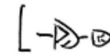
$\{(\begin{pmatrix}x_1\\x_2\end{pmatrix}, y) \mid \exists \hat{c}, \hat{d}\ x_1 \geq \hat{c} \wedge x_2 \geq \hat{d} \wedge \hat{c} + \hat{d} = y\} =$ 

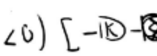
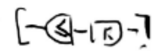
 $= \{(x, \bullet) \mid \exists c\ x \geq c\} \overset{\text{Tautologic}}{=} \{(x, \bullet) \mid x \in \mathbb{R}\} =$ 

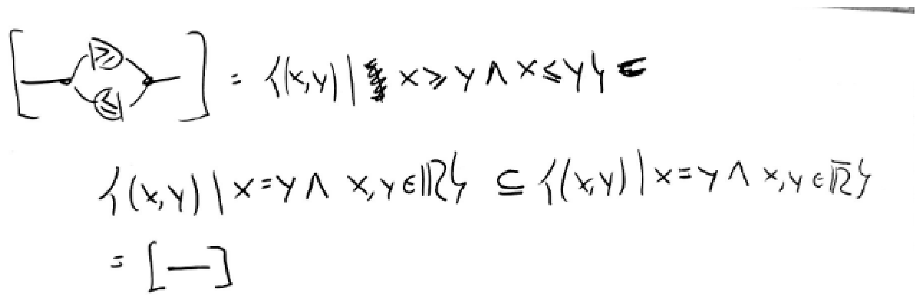
 $= \{(x, \bullet) \mid x = 0\} = \{(x, \bullet) \mid x \geq 0 \wedge x = 0\}$

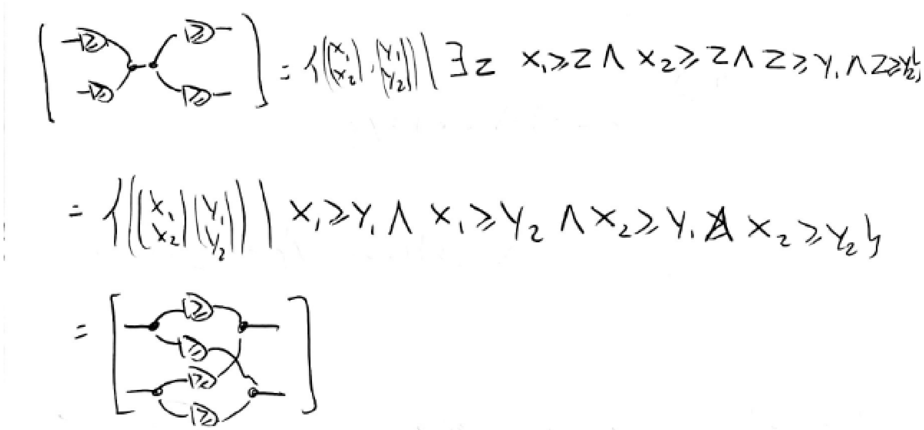
$\subseteq \{(x, \bullet) \mid x \geq 0\} =$ 

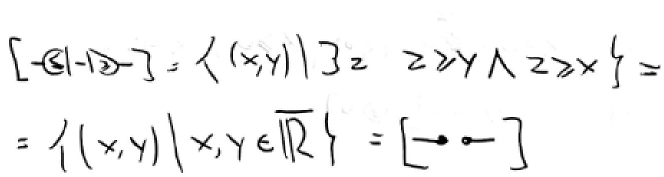
$(k>0)$  $= \{(x,y) \mid kx \geq y \wedge k > 0\} = \{(x,y) \mid x \geq \frac{y}{k} \wedge k > 0\}$

$= \{(x,y) \mid \exists c\ x \geq c \wedge kc = y \wedge k > 0\} =$ 

$k < 0)$  = Same as above, flip the $\geq$ = 



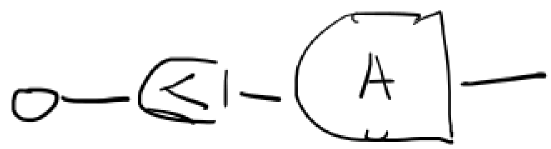




$\square$

Lemma 5 (Expressivity). *For every $f : m \to n$ in* **PCRel** *there exists a Σ-term $s : m \to n$ in $\mathrm{Hom}_{\mathrm{FreeP}_{\mathbf{GPA}_{\vdash}}}(m, n)$ such that $[s] = f$.*

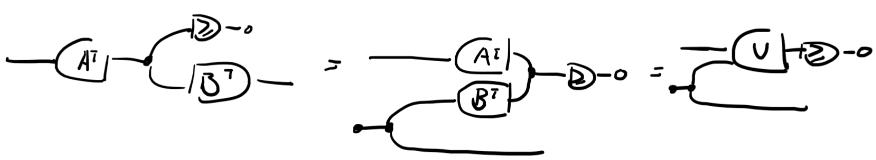
Proof. By Definition 22, every polyhedral cone has the form $\{x \mid Ax \geq 0,\ x \in \mathbb{R}^n\}$. This is precisely the interpretation of the diagram:



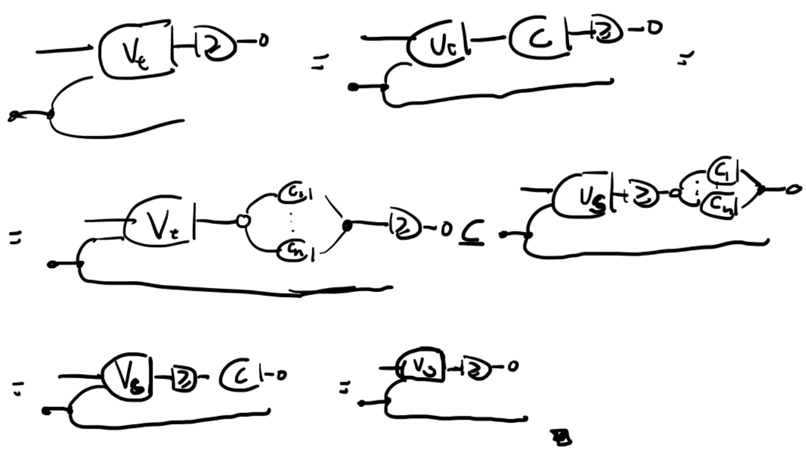
□

Lemma 6 (Completeness)**.** *If* $[s] = [t]$ *in* **PCRel***, then* $s = t$ *in* $\mathrm{FreeP}_{\mathbf{GPA}_{\vdash}}$.

Proof. By the second normal form, we write s and t in V-representation form:



The semantics of such a diagram is the conic hull of the column vectors of V. Since $[s] = [t]$ implies in particular $[s] \subseteq [t]$, the conic combinations of V_t are a subset of those of V_s. Hence there exists a non-negative matrix C such that $CV_t = V_s$. The diagrammatic witness is:



The inclusion $[s] \subseteq [t]$ follows by the same argument, yielding equality. □

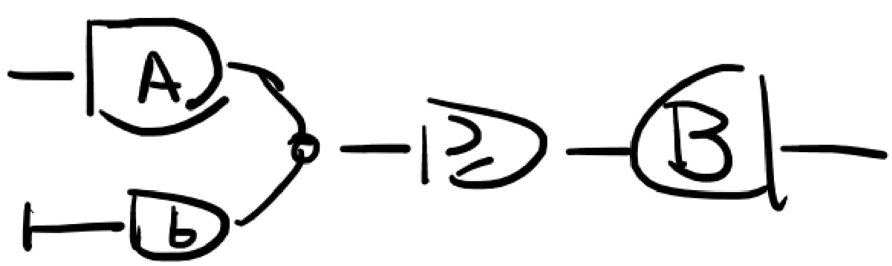
Theorem 5. **PCRel** *is presented by* $\mathrm{FreeP}_{\mathbf{GPA}_{\vdash}}$.

Proof. Soundness, expressivity, and completeness were established by the preceding three lemmas. Together they imply that the canonical PROP morphism is an isomorphism. □

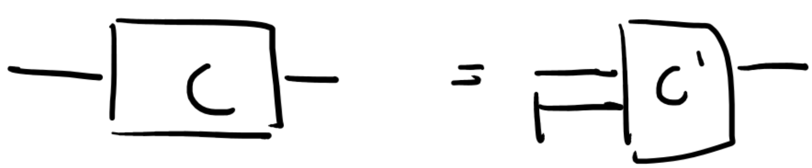
5.2 The algebra GPA and PolyRel

We now extend the theory from polyhedral cones to general polyhedra by reintroducing the affine generator $\vdash$. Geometrically, this corresponds to allowing translations: a polyhedron is a translated polyhedral cone, and the generator $\vdash$ provides the syntactic means to encode the right-hand side vector b in $Ax \geq b$. The normal form, soundness, and expressivity arguments all extend smoothly from the homogeneous case.

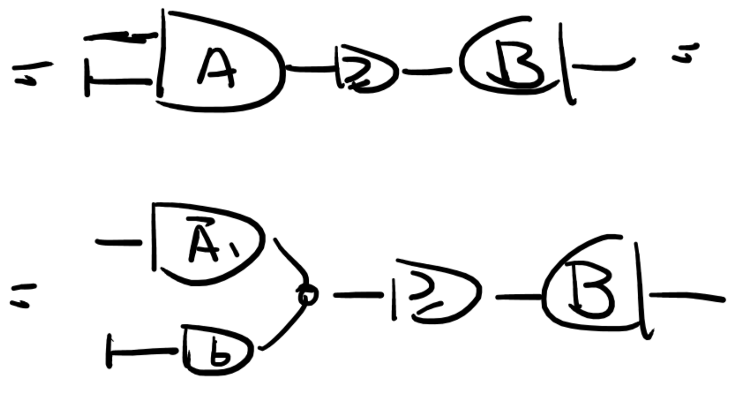
Theorem 6 (Normal Form)**.** *For every diagram* c *in* **GPA***, there exists a diagram* d *such that* $c = d$ *modulo the axioms and* d *is of the form:*



Proof. The derivation proceeds directly. Let c be a diagram of **GPA**. Then we can collect all $\vdash$ on the left such that c' is a diagram of $\mathbf{GPA}_{\vdash}$



From 4 we know that c' admits a normal form in terms of matrices A and B. Breaking down the matrix A in columns we get:



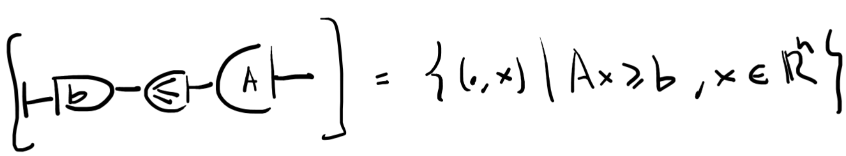
thus the thesis. $\square$

Lemma 7 (Soundness)**.** *If $s = t$ in* $\mathrm{FreeP}_{\mathbf{GPA}}$*, then $[s] = [t]$ in* **PolyRel***.*

Proof. This follows immediately from the soundness of **PCRel**. Since the generator $\vdash$ introduces no new axioms beyond those already verified for **PCRel** and **GAA**, and both of those theories are sound in **PolyRel**, soundness carries over directly. $\square$

Lemma 8 (Expressivity)**.** *For every $f : m \to n$ in* **PolyRel** *there exists a Σ-term $s : m \to n$ in $\mathrm{Hom}_{\mathrm{FreeP}_{\mathbf{GPA}}}(m, n)$ such that $[s] = f$.*

Proof. By Definition 23, every polyhedron has the form $\{x \mid Ax \geq b\}$. Appealing to this representation directly, the polyhedron is expressed as the interpretation of the following diagram, where the generator $\vdash$ encodes the right-hand side vector b:



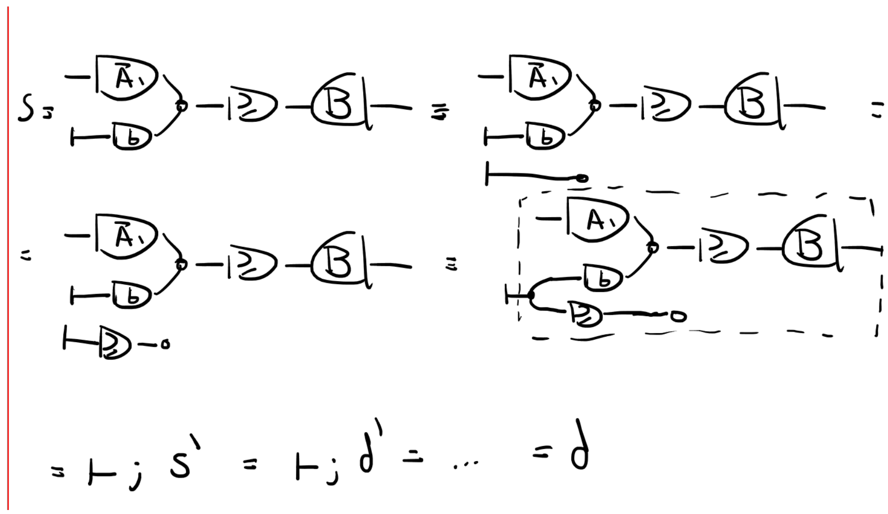
$\square$

Lemma 9 (Completeness)**.** *If $[s] = [t]$ in* **PolyRel***, then $s = t$ in* $\mathrm{FreeP}_{\mathbf{GPA}}$*.*

Proof. Every polyhedron P can be homogenised by introducing a slack variable, reducing the affine case to the conic one $P^H = \{(x, y) \in k^{n+1} | Ax + by \geq 0, y \geq 0\}$ which admits a normal form 4. Under this homogenisation, for each two diagrams $s, t \in$ **GPA** with $[s] = [t]$ in **PolyRel**, there are s', t' in $\mathbf{GPA}_{\vdash}$ such that $[s'] = [s]^H, [t'] = [t]^H$. Since two polyhedrals P, Q are equal iff $P^H = Q^H$, then

s and t have equal semantics in **PCRel**. Completeness of $\mathbf{GPA}_{\vdash}$ then gives equality of the homogenised diagrams, and de-homogenising recovers equality of s and t in $\mathrm{FreeP}_{\mathbf{GPA}}$.

Diagrammatically this goes as follows:



Whereas the interpretation of the diagram in the dotted box is $\{(x, y, z) \in k^m | Ax + by \geq Bz, y \geq 0\}$, which is the same as P^H. Therefore, the semantics of s' is that of the homogenisation of $[s]$ and follows the above argument. □

Theorem 7. **PolyRel** *is presented by* $\mathrm{FreeP}_{\mathbf{GPA}}$.

Proof. By the preceding three lemmas. □

6 The Linear Programming Category

The two graphical theories developed so far have each extended the categorical framework by adding a new class of morphisms. We now introduce the final and most expressive theory of the thesis: a graphical algebra for parametric linear programs. Linear programs combine both the inequality structure of polyhedra and the notion of an objective function, and their value functions are precisely the polyhedral convex functions. The central result of this section is the presentation of a complete monoidal theory for right-hand side parametric linear programming.

6.1 Preliminary Results in Linear Programming

We begin with several classical results that establish the tight relationship between piecewise affine convex functions, polyhedral functions, and parametric linear programs. These will serve as the semantic foundation for the graphical algebra introduced in the following subsection.

Definition 30 (Piecewise Affine Convex Function)**.** *A function $f : \mathbb{R}^n \to \mathbb{R}$ is called* piecewise affine convex *if there exist finitely many affine functions $a_i^\top x + b_i$, $i = 1, \dots, k$, such that*

$$f(x) = \max_{i=1,\dots,k} \{a_i^\top x + b_i\}.$$

Definition 31 (Polyhedral Function)**.** *A function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is* polyhedral *if its epigraph*

$$\operatorname{epi}(f) := \{(x, t) \mid t \geq f(x)\}$$

is a polyhedron.

Definition 32 (RHS Parametric Linear Program)**.** *A* right-hand side parametric linear program *has value function*

$$v(b) := \inf_x \{c^\top x \mid Ax \geq b\}.$$

The following two propositions establish that all three notions above are equivalent descriptions of the same class of functions. This equivalence is the semantic cornerstone of the section: it is what allows us to treat polyhedral functions and parametric linear programs as morphisms in the same category.

Proposition 15. *A function is piecewise affine convex if and only if it is a polyhedral function.*

Proof. ($\Rightarrow$) Let $f(x) = \max_{i=1,\dots,k}(a_i^\top x + b_i)$. Then

$$\operatorname{epi}(f) = \bigcap_{i=1}^{k} \{(x, t) \mid t \geq a_i^\top x + b_i\},$$

which is an intersection of finitely many halfspaces, hence a polyhedron.

($\Leftarrow$) If f is polyhedral, its epigraph is a polyhedron, so there exist finitely many affine inequalities $t \geq a_i^\top x + b_i$ such that $f(x) = \max_{i=1,\dots,k}(a_i^\top x + b_i)$. Thus f is piecewise affine and convex. □

Proposition 16. *A function $v : \mathbb{R}^m \to \mathbb{R} \cup \{+\infty\}$ is piecewise affine convex if and only if it is the value function of a right-hand side parametric linear program.*

Proof. ($\Rightarrow$) Let $v(b) = \max_{i=1,\dots,k}(y^i)^\top b$ and define $Y := \operatorname{conv}\{y^1, \dots, y^k\}$. Then $v(b) = \max_{y \in Y} b^\top y$. Since Y is a polyhedron admitting the representation $Y = \{y \mid A^\top y = c,\ y \geq 0\}$, linear programming duality gives $v(b) = \inf\{c^\top x \mid Ax \geq b\}$.

($\Leftarrow$) Let $v(b) = \inf\{c^\top x \mid Ax \geq b\}$. Dualising, $v(b) = \sup\{b^\top y \mid A^\top y = c,\ y \geq 0\}$. Since the feasible set $Y = \{y \mid A^\top y = c,\ y \geq 0\}$ is a polyhedron, the supremum is attained at its extreme points $y^1, \dots, y^k$, yielding $v(b) = \max_{i=1,\dots,k}(y^i)^\top b$. Thus v is piecewise affine convex. □

The following proposition is the key technical lemma for the completeness proof of **GLP**. It states that two parametric linear programs with the same value function must share the same epigraph polyhedron, providing the bridge between semantic equality and polyhedral equality that the completeness argument requires.

Proposition 17. *Let*

$$v_i(y) := \inf\{\mathbf{1}^\top x \mid A_i x \geq B_i y\}, \qquad i = 1, 2,$$

where $A_i \in \mathbb{R}^{m_i \times n_i}$ and $B_i \in \mathbb{R}^{m_i \times k}$. If $v_1(y) = v_2(y)$ for all $y \in \mathbb{R}^k$, then both programs admit a common representation through the same epigraph polyhedron:

$$v_i(y) = \inf\{t \mid (y, t) \in E\}, \qquad E := \mathrm{epi}(v_1) = \mathrm{epi}(v_2).$$

Proof. For each $i = 1, 2$, the epigraph of v_i satisfies

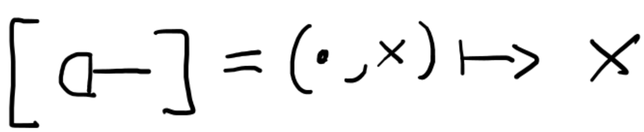
$$(y, t) \in \mathrm{epi}(v_i) \iff \exists x : A_i x \geq B_i y,\ \mathbf{1}^\top x \leq t,$$

so $\mathrm{epi}(v_i)$ is the projection onto (y, t) of the polyhedron $P_i = \{(x, y, t) \mid A_i x \geq B_i y,\ \mathbf{1}^\top x \leq t\}$. Projections of polyhedra are polyhedra, so each $\mathrm{epi}(v_i)$ is a polyhedron. Since $v_1 = v_2$ pointwise, we have $\mathrm{epi}(v_1) = \mathrm{epi}(v_2) =: E$. Setting $E = \{(y, t) \mid Cy + dt \geq 0\}$, we obtain $v_i(y) = \inf\{t \mid Cy + dt \geq 0\}$ for both $i = 1, 2$. □

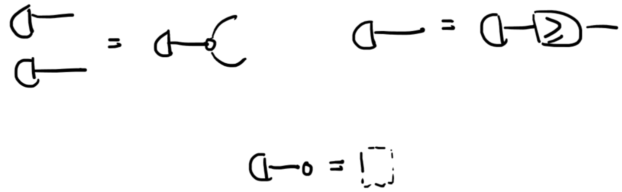
6.2 Graphical Algebra for Parametric Linear Programs

With the semantic equivalences established, we now introduce the final graphical algebra of the thesis. The algebra **GLP** extends **GPA** with a single new generator whose semantics encodes the linear objective function $c^\top x$. Just as the gray arrow in **GQA** lifted the theory from binary relations to weighted relations by "measuring" the ℓ_2 norm, this new generator lifts **GPA** from polyhedral constraints to full linear programs by attaching a linear cost to the feasible region.

Definition 33 (**GLP**)**.** *We call* **GLP** *the symmetric monoidal theory obtained by extending* **GPA** *with the following generator:*



together with the following axioms governing its interaction with the existing generators:



The axioms represent expected properties of the linear cost function. The one on the left express the linear property $f(x) + f(y) = f(x + y)$. Observe that f is the interpretation of the ᗡ − and the sum denotes the vertical composition; composing two ᗡ − vertically (adding) is the same as adding the wires in a single ᗡ −. In turn, the equation on the left express the fundamental property of minimization, the infimum of any quantity lies in its lower bound.

$$\inf_x \{x | x \geq y\} = y$$

Finally, the axiom at the middle is the tautological case.

$$inf 0 = 0$$

Definition 34. *We call* $\mathrm{FreeP}_{\mathbf{GLP}}$ *the PROP freely generated by* **GLP**.

We now define the two equivalent semantic categories for this algebra. The first, **LPRel**, describes morphisms as parametric linear programs; the second, **PolyFun**, describes them as polyhedral functions. The equivalence between the two follows immediately from the propositions above, and we will use both descriptions interchangeably according to which is more convenient.

Definition 35 (The prop **LPRel** of Linear Programs)**.** *The prop* **LPRel** *is defined as follows:*

- *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space $\mathbb{R}^n$.*
- *A morphism $n \to m$ is a right-hand side parametric linear program with value function*
$$(x, y) \mapsto \inf_z \{c^\top z \mid Az \geq B[x \ y]^\top\},$$
where $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$.
- *Composition is given by minimisation over the shared variable.*
- *The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is direct sum.*

Definition 36 (The prop **PolyFun** of polyhedral functions)**.** *The prop* **PolyFun** *is defined as follows:*

- *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space $\mathbb{R}^n$.*

- *A morphism $n \to m$ is a polyhedral function $f : \mathbb{R}^{n+m} \to \overline{\mathbb{R}}$.*
- *Composition is given by infimal convolution over the shared variable.*
- *The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is $f \otimes g := f + g$.*

Remark 7. *The two categories are isomorphic as PROPs:* **PolyFun** $\cong$ **LPRel**. *This is a direct consequence of the equivalence between polyhedral functions and right-hand side parametric linear programs established in the propositions above. We will therefore use the two names interchangeably, choosing whichever description is more natural in a given context.*

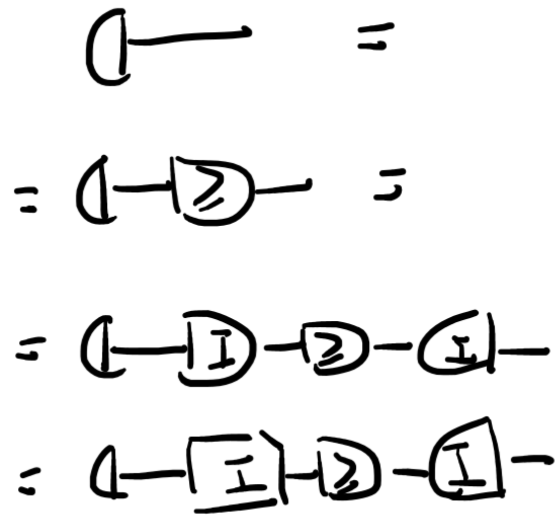
The normal form theorem for **GLP** extends the one for **GPA**: every diagram can be reduced to a **GPA** block encoding the constraints, composed with the new generator encoding the objective. The proof proceeds by structural induction, using the existing normal form for **GPA** as the inductive base.

Theorem 8 (Normal Form)**.** *For every diagram $c \in$ **GLP**, there exists a diagram c' in normal form such that $c = c'$:*

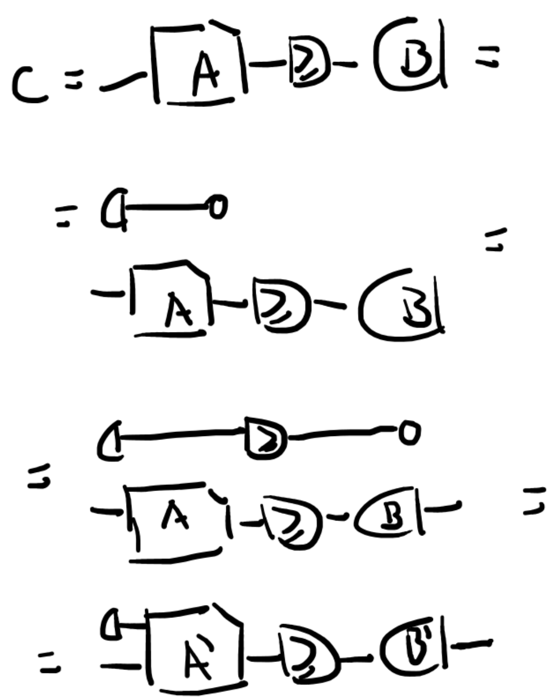
With A and be being an affine transformation. Namely:

Proof. The proof is by structural induction. First we show the two base cases:

- The generator ⊂ −



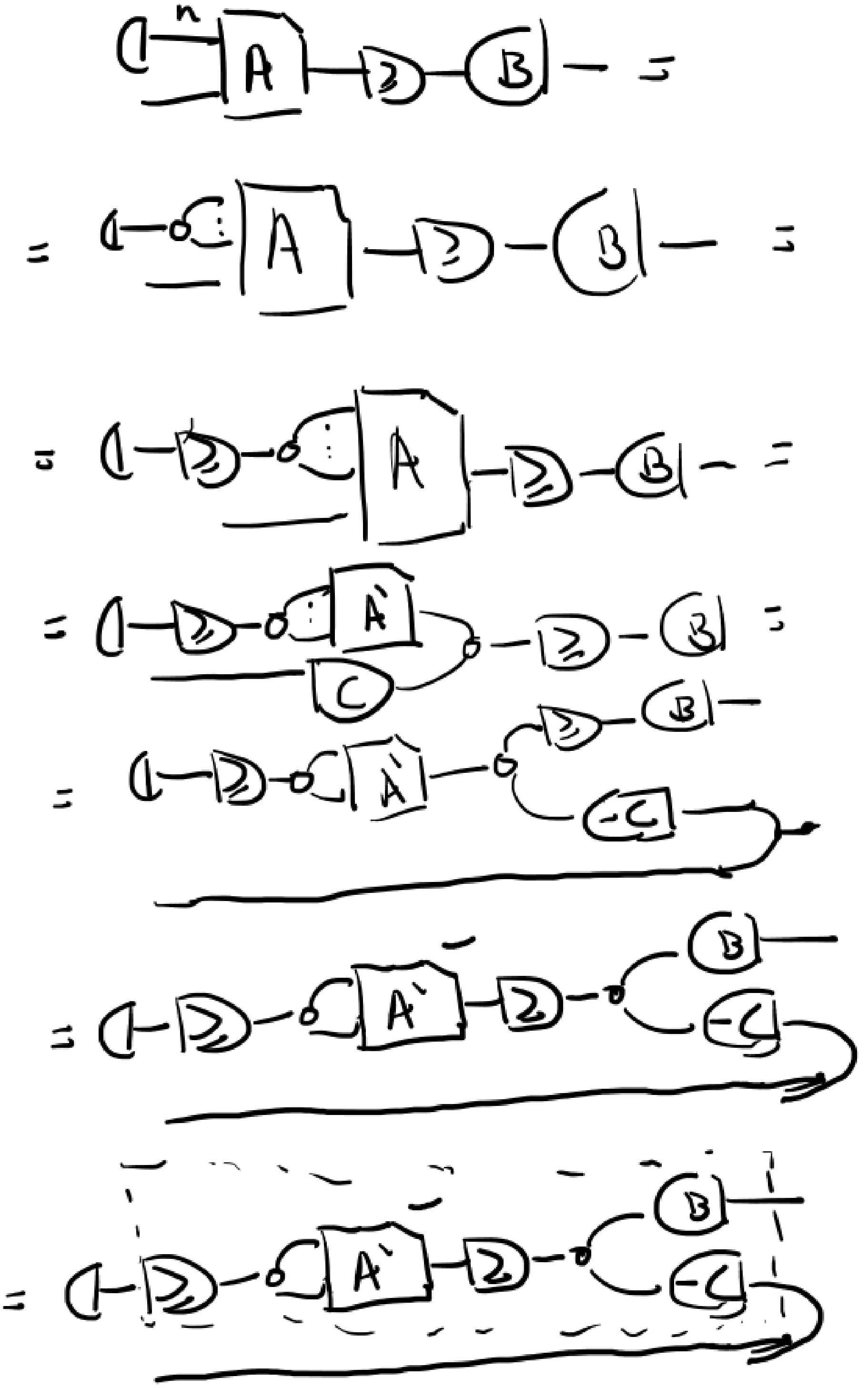
- A diagram c **GPA**



For the inductive step, we verify that for any two diagrams $c, d \in \mathbf{GLP}$ satisfying the inductive hypothesis, both horizontal and vertical composition preserve the normal form. Closure under vertical composition follows immediately from the normal form. For horizontal composition, we apply Fourier-Motzkin Elimination in the **GLP** part of the diagram to recover the normal form. □

The following corollary reveals the geometric meaning of the normal form: the **GPA** part of every **GLP** diagram encodes the epigraph of the corresponding value function. This is the key structural insight connecting the syntax to the semantic equivalence of Proposition 17.

Corollary 1 (Epigraph Form)**.** *Given a diagram $\mathsf{d} \in \mathbf{GLP}$ in normal form, it can always be rewritten as a diagram d' of the form:*


n
A
B
A`
C
-C

where the semantics of the dotted diagram is

$$\left\{(t, y) \mid \exists\, x,\ A'x + a \geq [B\ \ -C]\, y,\ t \geq x,\ y = \begin{bmatrix} y_1 \\ \\ y_2 \end{bmatrix}\right\}$$

. This is the epigraph of the original diagram d

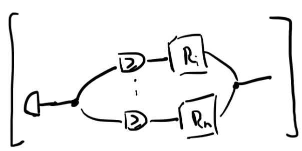
$$\begin{aligned} [d](y_1, y_2) &= \inf_z\ z \quad s.t.\ Az + a \geq By_1 - Cy_2 \\ \mathrm{epi}([d]) &= \{(t, y_1, y_2) \mid \exists\, z \leq t,\ Az + a \geq By_1 - Cy_2\} \end{aligned}$$

such that $A = [A'\ C]$*. We call this the* epigraph form *of a linear program.*

We are now ready to prove the three presentation properties for **GLP**.

Lemma 10 (Expressivity)**.** *For every* $f : m \to n$ *in* **PolyFun** *there exists a* Σ*-term* $s : m \to n$ *in* $\mathrm{Hom}_{\mathrm{FreeP}_{\mathbf{GLP}}}(m, n)$ *such that* $[s] = f$*.*

Proof. Since every polyhedral function is the maximum of a finite family of affine functions, it suffices to show that every piecewise affine convex function is expressible in **GLP**. For an arbitrary such function $f(y) = \max_i R_i(y)$, where $\{R_i\}_{i=1,\dots,n}$ is a family of affine functions, the following diagram provides the required syntactic representative:



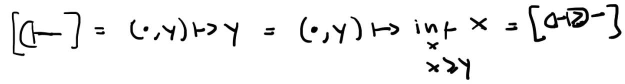
$$= \inf_x\ x + \{| x \geq R_i y |\}_{i=1,\dots n}$$

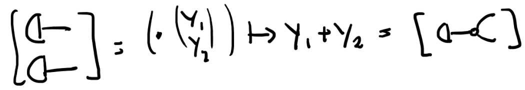
$$= \inf_{\substack{x \\ x \geq R_1 y \\ \vdots \\ x \geq R_n y}} x \;=\; \max_i \{R_i y\}_{i=1,\dots n}$$

□

Lemma 11 (Soundness)**.** *If* $s = t$ *in* $\mathrm{FreeP}_{\mathbf{GLP}}$*, then* $[s] = [t]$ *in* **PolyFun***.*

Proof. All axioms inherited from **GPA** are already sound by the results of the previous section. It suffices to verify that the new axioms involving the LP generator preserve the semantics, which follows by direct computation:





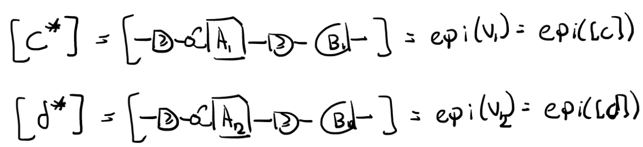
$\square$

The final result of the thesis is the completeness theorem for **GLP**. The proof brings together all the main tools developed throughout: the epigraph form of Corollary 1, the shared epigraph property of Proposition 17, the completeness of **GPA**, and the FME-based normal form. We state it as a conjecture because the completeness of the **GPA** polyhedra case, which the argument appeals to, still has a gap in the proof of Completeness for **PolyRel**; subject to that result, the argument below goes through.

Lemma 12 (Completeness)**.** *If* $[c] = [d]$ *in* **PolyFun**, *then* $c = d$ *in* $\mathrm{FreeP}_{\mathbf{GLP}}$.

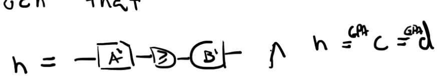
Proof. Let $v_1 = [c]$ and $v_2 = [d]$. By assumption $v_1 = v_2$, so Proposition 17 gives $\mathrm{epi}(v_1) = \mathrm{epi}(v_2) =: E$.

By Corollary 1, both c and d admit epigraph forms $c = ⊐ -; c^*$ and $d = ⊐ -; d^*$ with $[c^*] = [d^*] = E$ in **PolyRel**:



$$[c^*] = [d^*]$$

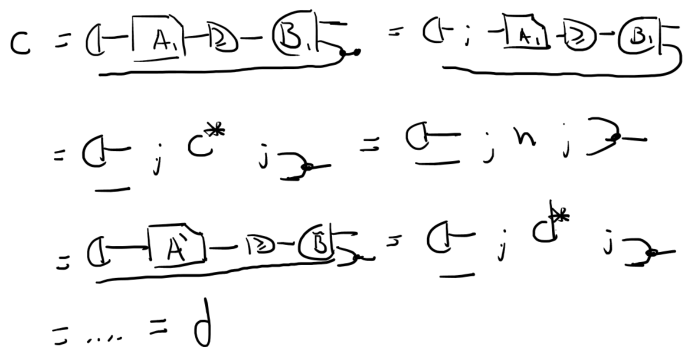
$\Rightarrow$ By completeness of GPA, $\exists n$ such that



Since $[c^*] = [d^*]$ in **PolyRel**, completeness of **GPA** gives $c^* = d^* = n$ in $\mathrm{FreeP}_{\mathbf{GPA}}$: Therefore,

$$c = ⊐ -; c^* = ⊐ -; n = ⊐ -; d^* = d,$$

as witnessed diagrammatically by:



□

As consequence of the three lemmas above

Theorem 9. **LPRel** *and* **PolyFun** *are both presented by* $\mathrm{FreeP}_{\mathbf{GLP}}$.

Proof. By the preceding three lemmas. □

Finally, we've built, through the addition of one generator and three axioms, an elegant and complete diagrammatical theory for Parametric Linear Programming. We argue that, such result, not just unlocks a new way to interact with LP problems in a algebraic and computational way, but also provide a new perspective to optimization through the lens of a compositional framework. Whereas optimization, namely LP problems, are lifted above the settings of functional thinking and achieves a new status as a relational-theoretic field embeded into the Extended-Lawvere Quantale category.

The **GLP** theory, in turn, has a natural extension to a even more expressive theory when enriched with the gray arrow generator from **GQA**. We believe that the class of quadratic programs could be, just as was done in this work, fully axiomatized by such broader SMIT. However, such studies are left for future works, along with other possible applications of current results.

7 Conclusion

This thesis begun with a philosophical claim, that the nature of a mathematical object is fully determined by its relationships to other objects, and that universal constructions reveal connections invisible to any particular viewpoint. Optimization, by contrast, is often presented as a collection of problem-specific techniques. The central claim of the thesis is that these two perspectives are not merely compatible, but that optimization *emerges* from categorical structure naturally, as an intrinsic property of relational composition and is natural to mathematical thinking.

To demonstrate that, we took advantage of the expressivity of graphical languages. The development followed a strict linear discipline, adding exactly one new generator at a time and verifying at each step that the resulting algebra is sound, expressive, and complete with respect to a precisely defined semantic category. This methodology is itself one of the work's implicit arguments. The fact that such a linear development is possible, that each new class of optimization problems requires only a single additional generator and a finite set of axioms, is evidence that the compositional structure is not imposed artificially but is genuinely present in the mathematics.

The following diagram illustrates the relationships among all the categorical structures developed in this thesis. Moving upward corresponds to increasing expressive power: each layer generalises the one below by enlarging the class of morphisms, while preserving the compositional structure. The morphisms marked $\hookrightarrow$ are inclusion functors, embedding each category as a full subcategory of the one above. The morphism marked $\mathbf{1}$ denotes the indicator function embedding, which sends a set-valued relation to its $\{0, +\infty\}$-valued characteristic function.

What this hierarchy makes visible is something that would be difficult to articulate within any single one of its layers. Polyhedral relations and quadratic relations are not merely two classes of optimization problems that happen to admit graphical representations; they are parallel enrichments of the same underlying affine structure, each adding a different kind of expressive power. Their common base in **AffRel** and their common embedding into **Cost-Relations** is not an accident of notation but a theorem, witnessed by the inclusion functors in the diagram. A single compositional framework within which relations, functions, constraints, and objectives are unified as morphisms, and within which optimization is not a technique applied to a problem but an algebraic operation itself.

The broader significance of this perspective lies in what it enables beyond the specific algebras treated here. The diagrammatic calculus established for **GLP** plays the same role for linear programming that Graphical Linear Algebra plays for matrices: it provides a language of normal forms, rewriting rules, and completeness theorems within which optimization arguments can be carried out symbolically, before and independently of any numerical computation. This opens several directions for future work. One of them is the exploration of the functoriality of the bifunction adjoint established in Section 3. Such observation suggests that a duality theory analogous to LP duality should be expressible diagrammatically in these settings as well.

A more applied direction concerns the computational implications of diagrammatic normal forms. If two optimization problems are equal as morphisms in **LPRel** if and only if their diagrams are related by the rewriting rules of **GLP**, then diagram simplification is problem simplification: a rewriting system on string diagrams is simultaneously a symbolic preprocessor for optimization solvers. Such an application of graphical algebras for theorem solvers was already pointed in other works [BDZ21; BDS21].

This thesis has taken the first step of the road to fully express optimization within it's own algebraical settings. It has shown that the language of string diagrams can absorb, unify, and extend three of the central theories of mathematical optimization within a single categorical framework built incrementally from the ground up. The theories here presented are not three separate pieces wearing categorical clothing, but a single theory, viewed at three different levels of expressive resolution.

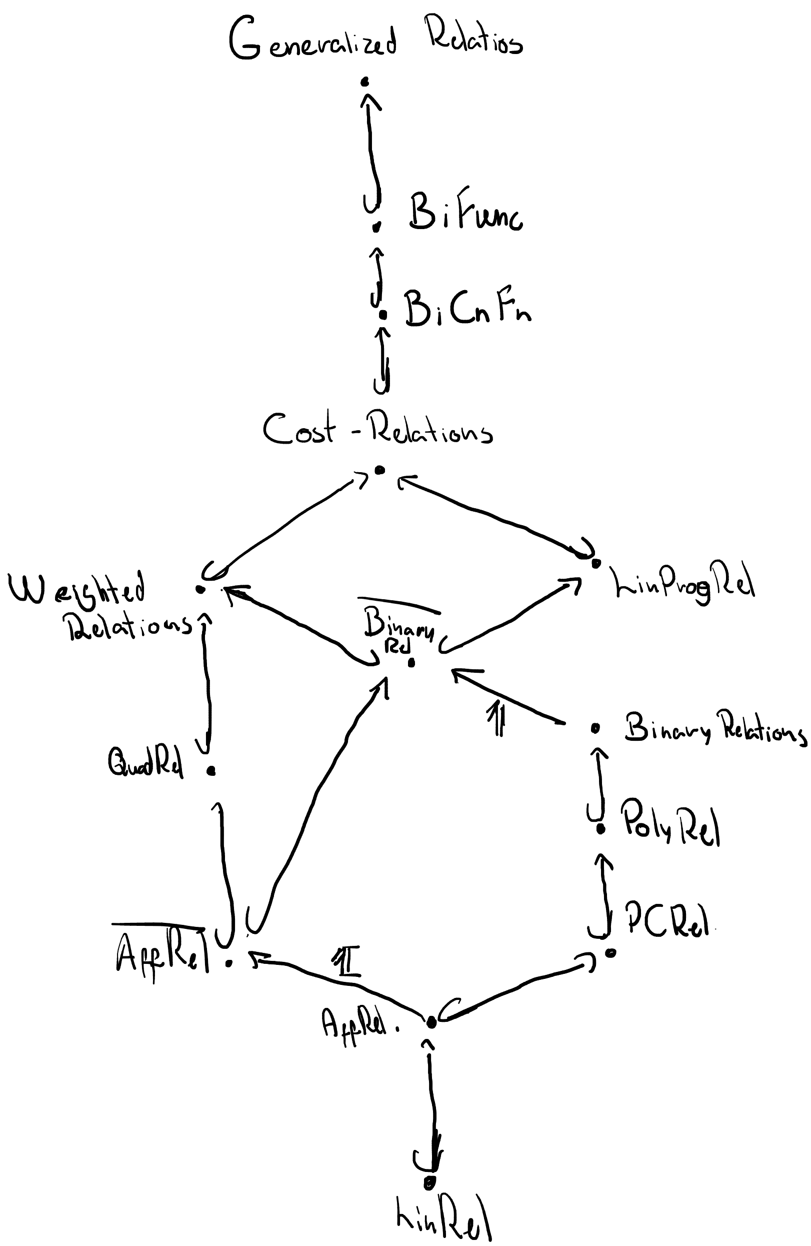


Figure 3: The hierarchy of relational categories developed in this thesis. Each inclusion preserves the monoidal and compositional structure of the layer below.

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